

UNIVERSAL DETERMINANT COMPLETION, EFFECTIVE PIERI MULTIPLICATION, AND INTRINSIC PRIMARY BOUNDARY LAWS IN MULTIPLICATION FAILURE

QIAN QI

ABSTRACT. We study the nonreduced Fitting scheme on a Grassmannian on which multiplication fails for cube-zero algebras with Hilbert function $(1, 4, 6)$. Its nilpotent filtration is intrinsic and recovers, on the first layer, the ramification quartic K3 surface with its polarization.

A determinant-completion theorem supplies the uniform algebraic bound behind the entire boundary theory. If V has dimension e , $r \geq 1$, and $\gamma : \text{Sym}^r V \rightarrow S$, then for the generic endomorphism M ,

$$(\det M)^{\binom{e+r-1}{r-1}} \in I_{\dim S}(\gamma \text{Sym}^r M).$$

We prove that any finite-presentation family with such a bounded principal determinant power admits a finite locally closed stratification on which every residual colon commutes with arbitrary base change, the positive graded layers are flat, and their geometric associated-point packets, embedded multiplicities, Hilbert polynomials, and multiplication ranks are constant. Thus the universal determinant bound and the intrinsic primary boundary atlas are two parts of one finite mechanism.

For the $(1, 4, 6)$ problem the bound is $d^5 \in J$. At projection corank two we determine the generic primary signatures of all mixed-kernel and pure-block rank types. The two nontrivial rank-drop slices are certified over their rational function fields: the four-line slice has factor

$$u_0 u_1 ((s+9)u_0^2 + (3s+6)u_0 u_1 - 7u_1^2),$$

with nonzero discriminant $9s^2 + 64s + 288$, while the two-line slice has factor $10u_0^2 + u_0 u_1 - 4u_1^2$ and an embedded vertex module of length three, computed after generic saturation. At projection corank three the determinant depth is five or six. At projection corank four it is exactly five: right $GL(V)$ -stability of the maximal-minor ideal, semisimple projection, and explicit Cauchy–Pieri bideterminant multiplication give $d^4 \in J$, while $d^3 \notin J$.

The resulting graded algebra specializes in finite families and is packaged canonically by deformation to the normal cone. The abstract nonreduced failure scheme continues to determine the polarized quartic K3 surface.

1. INTRODUCTION

A rank condition records where multiplication fails. The scheme structure of that rank condition records how it fails. In the cube-zero multiplication problem considered here this distinction is geometric: the reduction is a Schubert divisor, whereas successive nilpotent layers recover a polarized Jacobian K3 surface, contact divisors on a Segre quadric, vertex-primary corrections, and the higher-rank boundary on which those objects specialize.

The central point of this paper is that these layers are not artifacts of a chosen determinantal presentation. They are powers of the intrinsic nilradical of the failure scheme. The determinantal presentation is used to calculate them; the resulting graded schemes, their associated points, and their multiplication are intrinsic.

Date: September 23, 2026 (Revision 129).

2020 Mathematics Subject Classification. 14M12, 14J28, 13C40, 14B05, 14D20, 05E05.

Key words and phrases. Multiplication failure, Fitting scheme, determinant completion, Pieri multiplication, nilpotent filtration, primary specialization, quartic K3 surface.

The argument has two scales. The first is universal: determinant completion bounds the principal nilpotence exponent in every symmetric power. The second is relative: once such an exponent is bounded, the entire colon tower is finite and can be stratified simultaneously. The bounded principal-colon theorem proved below makes kernel formation commute with arbitrary base change before any associated-point argument is invoked. Relative assassins then turn the finite tower into geometric support packets with constant embedded multiplicities. For the $(1, 4, 6)$ problem this mechanism is supplemented by exact rational-function-field calculations on transverse stabilizer slices and by explicit boundary degenerations.

1.1. The multiplication failure scheme. Let V be a complex vector space of dimension $e \in \{3, 4\}$, put $p = e(e - 1)/2$, and let

$$R \subset \mathrm{Sym}^2 V$$

be an e -dimensional relation space. Set

$$S_R = \mathrm{Sym}^2 V / R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

with multiplication induced by the quotient $\mathrm{Sym}^2 V \twoheadrightarrow S_R$ and with $(V \oplus S_R)S_R = 0$. On

$$X_R = \mathrm{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle. For every $m \geq 2$ the multiplication map defines the same zeroth-Fitting scheme

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}]). \quad (1.1)$$

Its reduction is the Schubert divisor $\widehat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the invariant open G_e° used throughout, the web is basepoint-free and its Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \quad (1.2)$$

is smooth. For $e = 4$ it is a quartic K3 surface.

1.2. Intrinsic reconstruction. On the projection-corank-at-most-one locus let D_R be the restriction of $\widehat{D}_R$, let $\mathcal{N}_R$ be its nilradical, and let E_R be the annihilator scheme of the first nilpotent layer. The first main theorem is the reconstruction result proved in Section 5.

Theorem 1.1 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

Then

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en))$$

as graded algebras. In particular, for $e = 4$ the abstract failure scheme determines the polarized quartic K3 $(Y_R, \mathcal{O}_{Y_R}(1))$.

1.3. The intrinsic graded boundary packet. Write $\widehat{\mathcal{N}}_R$ for the nilradical of $\mathcal{O}_{\widehat{D}_R}$. The schemes

$$\mathcal{Z}_j(\widehat{D}_R) = V(\mathrm{Ann}(\widehat{\mathcal{N}}_R^j))$$

and the products among $\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1}$ are invariant under every abstract isomorphism of $\widehat{D}_R$. On a Schur chart the reduced equation is $d = \det T$ and

$$\mathcal{I}_{\widehat{D}_R} = dJ.$$

The residual-colon theorem of Section 9 identifies

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad W_j = V((d) + (J : d^{j-1})). \quad (1.3)$$

The coordinate expression changes from chart to chart; the modules and schemes in (1.3) do not.

Theorem 1.2 (Boundary structure). *For $R \in G_4^\circ$ the following hold.*

- (1) *On every projection-rank chart,*

$$d^5 \in J, \quad \widehat{\mathcal{N}}_R^6 = 0.$$

This is the $(e, r) = (4, 2)$ case of the universal determinant completion theorem, whose exponent $\binom{e+r-1}{r-1}$ is sharp in the universal class.

- (2) *The multiplication of the positive graded pieces is the canonical quotient multiplication*

$$\mathcal{O}_{W_i} \otimes \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}$$

with the expected conormal-line twist. No additional scalar multiplication constants occur.

- (3) *If the pure block A_H is injective at projection corank two, the mixed block has nine orbit types. On the generic coefficient stratum of every orbit, the complete nonzero graded supports through W_3 , their Hilbert functions, radicals, associated supports, and vertex-primary corrections are given in Theorems 6.4 and 11.2. Genericity is proved over rational function fields on transverse stabilizer slices.*

- (4) *If A_H drops rank at projection corank two, every generic rank type (a, b) has an explicit primary signature. Besides the Segre quadric, the only positive-dimensional generic supports that occur are reduced divisors of two or four lines in one ruling. The remaining layers are vertex-primary; in the $(a, b) = (1, 3)$ row a length-three embedded vertex accompanies the two ruling lines.*

- (5) *At projection corank three, $d^3 \notin J$ and $d^5 \in J$. The two possible depths are separated by the intrinsic membership condition $d^4 \in J$. The rank-drop passage in the representation argument is functorial, not inferred from semicontinuity.*

- (6) *At projection corank four,*

$$d^3 \notin J, \quad d^4 \in J.$$

The second membership is proved from the actual Cauchy–Pieri multiplication in $\mathbb{C}[\text{End}(V)]$: the product of the $(4, 4, 4, 0)$ - and (4) -components has nonzero $(4, 4, 4, 4)$ -projection. Since that line is spanned by d^4 , the local nilpotency index is exactly five.

Thus the phrase *boundary atlas* refers to intrinsic graded data, not to the chosen coefficient matrices. The Macaulay matrices form a finite presentation atlas. Any two such presentation atlases have a common refinement and compute the same intrinsic primary signature.

1.4. Primary specialization. The preceding finite calculations are promoted to family statements in Section 11. The key homological step is explicit. If $B = P_S/J$, $C_j = B/d^{j-1}B$, and $I_j = d^{j-1}B$, we first flatten B and C_j . The exact sequence $0 \rightarrow I_j \rightarrow B \rightarrow C_j \rightarrow 0$ then makes I_j flat, and $0 \rightarrow K_j \rightarrow P_S \rightarrow I_j \rightarrow 0$ remains exact after arbitrary base change. Only after the equality $(K_j)_s = (J_s : d_s^{j-1})$ has been established do we apply relative assassins. Constructibility of fibrewise associated points and Noetherian induction give finite geometric support packets; their embedded multiplicities are measured by the finite-length torsion modules at the associated generic points.

The same section writes explicit one-parameter degenerations of the mixed block. For example an off-Segre kernel point specializes to a Segre point; a secant kernel line specializes to a tangent or either ruling line; rank-two kernels specialize to the two rank-one types; and both rank-one types specialize to the zero mixed map. The graded schemes in the two fibres are computed, so

the specialization records the birth of vertex-primary and ruling components rather than merely changing orbit labels.

1.5. The simple-contact locus. The general atlas contains the earlier primary theorem as its split form. We retain the following statement because it is the geometric entry point to the collision analysis.

Theorem 1.3 (Corank-two primary structure on the simple-contact open). *There is a nonempty invariant open $G_4^\dagger \subset G_4^\circ$ such that, for each $R \in G_4^\dagger$, a dense open set of rank-two images $H \subset V$ has, after an étale extension and smooth auxiliary coordinates,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) \cap P_1^2 \cap P_2^2 \cap P_3^2 \cap P_4^2 \cap \mathfrak{m}^5. \quad (1.4)$$

The rank-two stratum is an additional associated support and $\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2$.

Repeated roots, the containment locus, the decomposable-kernel wall, and all remaining mixed-kernel orbits are treated without splitting the contact quartic.

1.6. Variation and finite ambiguity.

Theorem 1.4 (Variation and finite ambiguity). *For $e = 4$, the image of*

$$R \mapsto (Y_R, \mathcal{O}_{Y_R}(1))$$

in quartic K3 moduli has dimension nine. On a nonempty invariant open the induced map from projective equivalence classes of relation spaces to its image is generically finite.

The nine-dimensional count is classical Reye geometry, not a novelty claim. The manuscript-specific inverse statement is the recovery of the polarized K3 from an abstract nonreduced failure scheme. Revision 128 does not need an unproved generic-injectivity assertion: its broader structural consequence is instead the universal symmetric-power determinant-completion theorem.

1.7. Organization and novelty boundary. Section 2 fixes the relation model and Section 3 gives the Fitting presentation. Section 4 develops the intrinsic nilpotent filtration, followed by polarized reconstruction. Section 6 states the finite boundary theorems; the inherited corank-two and global residual calculations follow, and Section 10 proves the new determinant, genericity, and higher-corank statements. Section 11 proves the family and specialization results. Deformation to the normal cone is then used only as the standard package for the computed graded algebra. The moduli and classical/priority comparisons close the article.

Deformation to the normal cone, flattening stratification, Schur functors, and standard monomial straightening are classical tools. The new assertions are the specific residual Fitting algebra, the universal determinant-completion mechanism applied to it, effective Pieri nonvanishing for the corank-four membership, the generic and primary corank-two boundary tables, their specialization laws, and the reconstruction of the polarized K3 from the intrinsic nilpotent scheme. Ballico's 1993 paper remains cited as a documentary boundary; no theorem-level nonanticipation claim is made without its complete text.

2. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 2.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \mathrm{Sym}^2 \mathcal{H} \longrightarrow \mathrm{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\mathrm{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \mathrm{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition 2.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (1.2). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (2.1)$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e - 1$ and is the whole target. Intersecting $e - 1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma 2.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition 2.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension $p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective

incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$ when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma 2.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

3. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition 3.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\hat{D}} = (\det M)I_p(\gamma \mathrm{Sym}^2 M). \quad (3.1)$$

The reduction of $\hat{D}$ is the integral divisor $\hat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\hat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \mathrm{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (3.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\mathrm{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \text{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \text{Tot}_{P \times \mathbb{P}(S)} \text{Hom}(\mathcal{H}, S/\Lambda), \quad (3.2)$$

$$E \simeq \text{Tot}_{Y \times \mathbb{P}(S)} \text{Hom}(\mathcal{H}, S/\Lambda). \quad (3.3)$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem 3.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (3.4)$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \text{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\text{Sym}^2 H}$. Equation (2.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (3.1) is (t) . On Y , the restriction has rank $p - 1$ by Proposition 2.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (3.5)$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (3.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma 3.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\text{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \rightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \text{Spec } \mathcal{O}_{Z, \eta} \rightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

Corollary 3.4 (Ordinary powers). *For $n \geq 1$, write $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$. Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (3.6)$$

is an irredundant primary decomposition on U . With $\mathcal{T}_n = \text{tor}_{\eta_\Delta}(\mathcal{O}_{D^{[n]}})$,

$$0 \rightarrow \mathcal{T}_n = \mathcal{O}_{E^{[n]}}(-n\Delta) \rightarrow \mathcal{O}_{D^{[n]}} \rightarrow \mathcal{O}_{n\Delta} \rightarrow 0. \quad (3.7)$$

Its annihilator defines $E^{[n]}$. The generic multiplicity on Δ is n , the nilradical index along E is $2n$, and the length of $\mathcal{T}_n$ at η_E is $n(n+1)/2$.

Proof. In the local ring let $P = (t, f)$. The associated graded ring of P is a polynomial ring in two variables over $\mathcal{O}_E$, and the initial form of t is one of those variables. Hence $(P^{2n} : t^n) = P^n$, proving $(t^n) \cap P^{2n} = t^n P^n$. The same graded argument proves that these powers are primary. The witnesses t^n and f^{2n} at η_E prove irredundancy. Localizing at (t) contracts the ideal to (t^n) , so the generic torsion is $(t^n)/t^n P^n$, and cancellation gives its annihilator P^n . The class of t has index $2n$ at E , since t^{2n} belongs to the ideal but t^{2n-1} does not. At the generic point of Δ , f is a unit. Finally $\mathcal{O}_{U, \eta_E}/(t, f)^n$ has the $n(n+1)/2$ monomials of total degree less than n as successive graded basis elements. $\square$

Lemma 3.5 (Flat flags and arbitrary base change). *Let $U \rightarrow T$ be smooth, and let $E \subset \Delta \subset U$ be relative smooth closed subschemes, each inclusion a Cartier divisor. Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. Then $\mathcal{O}_U/P^n$ and $\mathcal{O}_U/L^n$ are flat over T . For $D_n = V(L^n P^n)$ the sequence*

$$0 \rightarrow L^n \otimes \mathcal{O}_{E^{[n]}} \rightarrow \mathcal{O}_{D_n} \rightarrow \mathcal{O}_U/L^n \rightarrow 0 \quad (3.8)$$

is exact and remains exact after any base change. Its base-changed terms are the indicated ordinary powers and products. In particular D_n is flat over T .

Proof. The immersion $E \subset U$ is regular of codimension two and has locally free conormal module. The conormal filtration of $\mathcal{O}_U/P^n$ has quotients $\text{Sym}^j(P/P^2)$ on E , for $0 \leq j < n$; these are base-flat. Induction proves flatness. The analogous principal filtration proves flatness of $\mathcal{O}_U/L^n$. Multiplication by a local equation t^n of L^n identifies the kernel in (3.8) with $L^n \otimes \mathcal{O}_{E^{[n]}}$, since t is a non-zero-divisor. Both outer terms are base-flat, so the middle term is base-flat. For a base algebra $A \rightarrow A'$, the only possible left obstruction in tensoring this short exact sequence is $\text{Tor}_1^A(\mathcal{O}_U/L^n, A')$, which is zero. The sequences defining L , P , and their powers stay exact for the same reason. Thus their images after base change are the actual ideals generated by the base-changed equations, and multiplication gives their stated products. No assertion about specialization of selected primary components is used. $\square$

Corollary 3.6. *Over G_e° , the schemes E , Δ , and U form the smooth relative flag of Lemma 3.5. All $D^{[n]}$ and $E^{[n]}$ are flat, and (3.7) commutes with arbitrary base change. Each complex fibre has (3.6).*

Proof. Use the universal quotient $\mathcal{S}$ and the graph bundles (3.2)–(3.3) over the smooth family of Proposition 2.3. The Fitting elimination works over the parameter ring, so it gives $t(t, f)$ relatively. Apply the lemma, and apply Corollary 3.4 separately on each fibre for the primary assertion. $\square$

4. THE INTRINSIC NILPOTENT-DEPTH FILTRATION

We isolate the functorial mechanism that unifies the two local models. Let Z be any noetherian scheme and let $\mathcal{N}_Z$ denote the nilradical of $\mathcal{O}_Z$. Since Z is noetherian, $\mathcal{N}_Z$ is a coherent nilpotent ideal. For $j \geq 1$ define

$$\mathcal{Z}_j(Z) = V(\text{Ann}_{\mathcal{O}_Z}(\mathcal{N}_Z^j)). \quad (4.1)$$

Lemma 4.1 (Functoriality). *Every isomorphism $f : Z \rightarrow Z'$ carries $\mathcal{Z}_j(Z)$ isomorphically to $\mathcal{Z}_j(Z')$ for every j . The construction commutes with restriction to open subsets.*

Proof. An isomorphism carries the nilradical to the nilradical, hence carries its j -th power and its annihilator ideal to the corresponding objects. Both formation of powers and annihilators commute with localization. $\square$

Apply this to $Z = \widehat{D}_R$. On the intrinsic open obtained by restricting to the smooth locus of Z_{red} , Proposition 3.1 identifies Z with D_R . Theorem 3.2 gives the local ring

$$A = \mathcal{O}/(t(t, f))$$

with nilradical generated by the class of t . Therefore

$$\text{Ann}_A((t)) = (t, f)/(t(t, f)),$$

so (4.1) recovers E_R scheme-theoretically.

Proposition 4.2 (Depth one is the polarized ramification support). *On the smooth-reduction locus of $\widehat{D}_R$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} = \mathcal{O}_{E_R}(-\widehat{\Delta}_R).$$

Consequently the pair

$$\left(\mathcal{Z}_1(\widehat{D}_R), \omega_{\mathcal{Z}_1(\widehat{D}_R)} \otimes (\widehat{\mathcal{N}}_R|_{\mathcal{Z}_1(\widehat{D}_R)})^{-9} \right)$$

is intrinsic when $e = 4$, and its graded section ring is the fourth Veronese of the polarized quartic section ring.

Proof. Only the last assertion is new notation. The annihilator calculation above is Theorem 3.2; the line-bundle identity and section-ring reconstruction are Proposition 5.2 and Theorem 1.1. $\square$

The point of (4.1) is that it makes no reference to a projection chart. The closed schemes are automatically nested:

$$\widehat{\mathcal{N}}_R^{j+1} \subset \widehat{\mathcal{N}}_R^j \implies \text{Ann}(\widehat{\mathcal{N}}_R^j) \subset \text{Ann}(\widehat{\mathcal{N}}_R^{j+1}), \quad \mathcal{Z}_{j+1} \subset \mathcal{Z}_j.$$

The new issue is therefore not formal functoriality but the scheme-theoretic computation of the layers as one crosses projection rank. Theorem 9.1 solves this globally: on the whole Grassmannian the residual colon

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R})$$

determines every annihilator and every associated-graded piece, including all projection coranks and rank-drop loci. The remaining task is geometric identification of those canonical colon schemes.

On the first higher-corank chart the next power is nonzero. The calculation in §7 gives, on the simple-contact open,

$$\widehat{\mathcal{N}}_R^3 = 0, \quad \text{Ann}(\widehat{\mathcal{N}}_R^2) = \mathcal{I}_{\Sigma_R}/\mathcal{I}_{\widehat{D}_R}.$$

Section 8 removes the simple-root assumption. Section 9 computes the two layers simultaneously across the corank-one/corank-two boundary. On the mixed-regular neighbourhood,

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_R}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}),$$

where $\mathcal{C}_R$ is the unsplit contact cone and $\mathcal{L}$ is the Schubert conormal line. In particular $\mathcal{Z}_1$ specializes from the ramification support to its canonical contact closure with the vertex thickening forced by the Fitting algebra, while $\mathcal{Z}_2$ is the rank-two vertex. On the first tensor-rank boundary of the mixed map, $\mathcal{Z}_2$ acquires a canonical length-three nonreduced structure (Theorem 9.5).

Thus

$$\mathcal{Z}_1 \rightsquigarrow \mathcal{Z}_2$$

is not merely a comparison of two calculations on disjoint strata: it is the associated-graded specialization of one intrinsic nilpotent algebra.

Remark 4.3 (Relation with canonical filtrations on multiple structures). Nilpotent and multiple structures have long been studied through canonical filtrations and associated graded modules; ribbons are the multiplicity-two prototype, and primitive multiple schemes carry a canonical chain of lower multiplicities. See [4, 15, 11]. The construction (4.1) is deliberately more elementary: it is available for the present Fitting schemes without assuming that they are primitive, locally complete-intersection multiple structures, or Cohen–Macaulay along every higher-corank stratum. The additional content here is that consecutive annihilator layers and their associated graded

modules have a specific geometric meaning forced by multiplication: the first carries the polarized ramification K3, the second detects the rank-two Schubert support, and their specialization through contact collisions is controlled by the unsplit contact Cartier ideal rather than by a choice of primary roots.

5. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (3.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \text{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma 5.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \text{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \text{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (5.1)$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\text{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\text{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\text{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\text{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\text{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every translation of $\text{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (5.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition 5.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (5.2)$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (5.3)$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_{\Delta}|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem 3.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (5.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem 1.1. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition 5.2, Lemma 5.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition 3.1 recovers D from $\widehat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark 5.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\text{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark 5.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

6. UNIVERSAL DETERMINANT COMPLETION AND THE INTRINSIC BOUNDARY ATLAS

The residual-colon filtration of Theorem 9.1 is intrinsic at every projection rank. We first isolate the determinant-completion mechanism in arbitrary dimension and symmetric degree. We then compute the projection-corank-two boundary on genuine generic coefficient strata, including all pure-block rank drops, and finally analyze projection coranks three and four. The finite coefficient matrices used below form a presentation atlas; the schemes W_j , their associated points, generic lengths, and graded multiplication form the intrinsic boundary packet.

Keep the notation of Theorem 8.1. Thus, at a point of projection rank r , put

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad W = V/H, \quad c = \dim W = 4 - r,$$

choose a Schur chart

$$M = I_H \oplus T, \quad d = \det T,$$

and write

$$\mathcal{I}_{\widehat{D}_R} = dJ, \quad J = I_6 \left[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T \right]. \quad (6.1)$$

The local nilradical is generated by the class of d .

Theorem 6.1 (Universal symmetric-power determinant completion). *Let V be a complex vector space of dimension e , let $r \geq 1$, and let*

$$\gamma : \text{Sym}^r V \rightarrow S$$

be a surjection, with $s = \dim S$. On $\text{End}(V)$ let M be the generic endomorphism, put $d = \det M$, and set

$$J_{\gamma, r} = I_s(\gamma \text{Sym}^r M).$$

Then

$$d^{\kappa(e,r)} \in J_{\gamma,r}, \quad \kappa(e,r) = \binom{e+r-1}{r-1}. \quad (6.2)$$

The exponent is sharp in the universal class: if $S = \operatorname{Sym}^r V$ and γ is an isomorphism, then $J_{\gamma,r} = (d^{\kappa(e,r)})$.

For $(e,r) = (4,2)$, $\kappa(4,2) = 5$. Thus the global depth bound below is a specialization of a dimension-free determinant-completion principle, rather than an isolated identity for the Hilbert function $(1,4,6)$.

Theorem 6.2 (Finite all-corank depth and multiplication). *For every $R \in G_4^\circ$ and every projection-rank chart,*

$$d^5 \in J. \quad (6.3)$$

Consequently

$$\widehat{\mathcal{N}}_R^6 = 0$$

on the whole Grassmannian. In particular

$$W_j = \emptyset \quad (j \geq 6),$$

and the special fibre of the intrinsic normal-cone specialization has at most five positive graded layers.

For $i, j \geq 1$, $i + j \leq 5$, the multiplication map

$$\frac{\widehat{\mathcal{N}}^i}{\widehat{\mathcal{N}}^{i+1}} \otimes \frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \longrightarrow \frac{\widehat{\mathcal{N}}^{i+j}}{\widehat{\mathcal{N}}^{i+j+1}}$$

is, under the identifications

$$\frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

the tensor twist of the canonical quotient multiplication

$$\mathcal{O}_{W_i} \otimes_{\widehat{\Delta}_R} \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}. \quad (6.4)$$

In particular it is surjective. Thus no additional multiplication constant is hidden after the schemes W_j are known.

We next sharpen the corank-two analysis. Assume A_H is injective. Put

$$E_H = S_R / \operatorname{im} A_H, \quad \beta_H = \overline{B}_H : H \otimes W \longrightarrow E_H, \quad \chi_H = \overline{C}_H : \operatorname{Sym}^2 W \longrightarrow E_H.$$

Here $\dim H = \dim W = 2$ and $\dim E_H = 3$. Let

$$Q = \mathbb{P}(H \otimes W)_{\operatorname{rk}=1} \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

be the Segre quadric.

Definition 6.3 (Mixed-kernel orbit type). The *mixed-kernel orbit type* of (R, H) is the rank $b = \operatorname{rank} \beta_H$, together with the incidence type of $\mathbb{P}(\ker \beta_H)$ with Q .

For $b = 3$ this is a point off Q or on Q . For $b = 2$ it is a secant line, a tangent line, an H -ruling $\mathbb{P}(h_0 \otimes W)$, or a W -ruling $\mathbb{P}(H \otimes w_0)$ contained in Q . For $b = 1$, the one-dimensional image of $\beta_H^* : E_H^* \rightarrow (H \otimes W)^*$ is generated by a tensor of rank two or rank one; equivalently the kernel plane cuts Q in a smooth conic or in a pair of ruling lines. The final type is $b = 0$.

These are exactly the nine $GL(H) \times GL(W) \times GL(E_H)$ -orbits for the mixed block. The first two are the mixed-regular and decomposable-kernel cases already encountered in Section 9; the remaining seven are higher-dimensional mixed kernels or failures of mixed surjectivity.

For a homogeneous ideal $K \subset P = \mathbb{C}[a, b, c, d]$, write

$$H_{P/K}(n) = \dim_{\mathbb{C}}(P/K)_n.$$

For each orbit, the *generic coefficient stratum* is the dense open obtained from a transverse stabilizer slice by deleting the explicit denominator and leading-coefficient divisors in Theorem 11.1. Equivalently it is the locus on which the rational-function-field Groebner bases used there retain their initial ideals. The isolated integral matrices listed in the proof are base points of those slices; they are not used as a substitute for genericity.

Theorem 6.4 (Corank-two mixed-kernel atlas). *Assume A_H is injective. Then the first positive graded support is determined without any mixed-regularity assumption:*

$$\mathcal{O}_{W_1} \simeq P/(\delta, g_0, \dots, g_4), \quad \delta = ad - bc, \quad (6.5)$$

where, on $P/(\delta)$ with $T = uv^t$, the degree-four classes span

$$h_H(u) \operatorname{Sym}^4(\mathbb{C}_v^2).$$

Thus, if $h_H \neq 0$,

$$\operatorname{Proj} W_1 = V(h_H) \times \mathbb{P}_v^1 \subset Q,$$

whereas $h_H = 0$ gives $W_1 = V(\delta)$.

On the generic coefficient stratum of each mixed-kernel orbit, the remaining graded supports and the exact local nilpotency index are as follows.

b	mixed-kernel type	index	W_2	W_3
3	point off Q	3	length 1	empty
3	point on Q	3	length 3, $H = (1, 2)$	empty
2	secant line	3	length 8, $H = (1, 4, 3)$	empty
2	tangent line	3	length 8, $H = (1, 4, 3)$	empty
2	H -ruling $\mathbb{P}(h_0 \otimes W)$	4	length 15, $H = (1, 4, 6, 4)$	length 1
2	W -ruling $\mathbb{P}(H \otimes w_0)$	3	$H(n) = n + 1$ for $n \geq 4$	empty
1	dual tensor rank 2	4	$H(n) = 2n + 2$ for $n \geq 3$	length 1
1	dual tensor rank 1	4	$H(n) = 3n + 3$ for $n \geq 3$	length 3
0	zero mixed map	4	$V(\delta)$	$V(\delta)$

For $b = 0$, surjectivity of the original quotient multiplication forces χ_H to be an isomorphism and

$$J = (\delta^3); \quad (6.6)$$

hence the last row is an exact identity, not only a generic one.

In the H -ruling and $b \leq 1$ rows one has $h_H = 0$. On the W -ruling row the contact quartic is nonzero on the transverse open and has the root forced by w_0 . For $b = 1$, the projectivization of W_2 has degree two in the dual-rank-two case and degree three in the dual-rank-one case. The length-one and length-three third layers are respectively the reduced rank-two vertex and the length-three vertex thickening already appearing on the decomposable-kernel wall.

The complement of the generic coefficient stratum is itself part of the atlas, rather than an omitted exceptional set. We make this finite refinement explicit.

Definition 6.5 (Macaulay valuation atlas). Let $P_c = \mathbb{C}[t_{ij} : 1 \leq i, j \leq c]$, $d = \det T$, and let J be the residual ideal on a fixed projection-rank chart. For $1 \leq k \leq 5$, let

$$M_k(J)$$

be the coefficient matrix whose columns are all products

$$m q, \quad q \in \mathcal{G}(J), \quad \deg m + \deg q = ck,$$

written in the monomial basis of $(P_c)_{ck}$, where $\mathcal{G}(J)$ is the displayed set of maximal minors in (6.1). Let v_k be the coefficient vector of d^k . Define

$$\varepsilon_k(J) = \text{rank}[\mathbf{M}_k(J) \mid v_k] - \text{rank } \mathbf{M}_k(J) \in \{0, 1\}.$$

Then

$$\varepsilon_k(J) = 0 \iff d^k \in J.$$

The locally closed loci obtained by fixing the ranks of the matrices $\mathbf{M}_k$, their augmented matrices, and the analogous matrices for the kernels

$$K_j = (J : d^{j-1}), \quad 1 \leq j \leq 5,$$

are called the *Macaulay valuation atlas*. All of their equations are minors of explicit finite coefficient matrices.

Theorem 6.6 (Bounded principal-colon primary stratification). *Let S be noetherian, let A be a finitely presented S -algebra which is flat over S , let $J \subset A$ be a finitely generated ideal, and let $f \in A$. Assume that $f^N \in J$ for some $N \geq 1$. Put*

$$B = A/J, \quad K_q = (J : f^q), \quad M_q = A/(f, K_q), \quad 0 \leq q < N.$$

Then S admits a finite locally closed stratification $S = \coprod_{\alpha} S_{\alpha}$ with the following properties.

For every α , all the assertions below are preserved by an arbitrary base change $T \rightarrow S_{\alpha}$.

- (1) *Formation of J and of every K_q commutes with base change. The modules*

$$B, \quad C_q = A/(J, f^q), \quad I_q = f^q B, \quad M_q$$

are flat over S_{α} .

- (2) *For each q there are finitely many closed relative support packets $Z_{q,\nu} \subset \text{Spec } A_{S_{\alpha}}$, flat after a further refinement, such that for every geometric point $\bar{s} \rightarrow S_{\alpha}$ the associated points of $(M_q)_{\bar{s}}$ are exactly the generic points of the irreducible components of the nonempty $(Z_{q,\nu})_{\bar{s}}$. Geometric splitting of a packet is allowed. Dimensions, incidence, and minimal-versus-embedded status are constant on the stratum.*

- (3) *If p is one of these geometric associated points, the canonical embedded multiplicity*

$$\mu_p(M_q) = \text{length}_{\mathcal{O}_{\text{Spec } A_{\bar{s},p}}} H_{p\mathcal{O}_{\text{Spec } A_{\bar{s},p}}}^0((M_q)_p)$$

is constant along the corresponding packet.

- (4) *Any prescribed finite collection of maps between the M_q may be included in the same refinement so that the relevant source, target, image and cokernel modules are flat; in particular their fibre ranks commute with base change.*

Thus a bounded principal power converts the a priori infinite primary problem into a finite, base-change-compatible invariant.

The theorem is independent of the special (1, 4, 6) model. In the present paper the universal determinant-completion theorem supplies $N = 5$ on every projection-rank chart. Consequently the residual colon algebra, its associated-point packets, and all graded multiplication maps admit one finite primary stratification before any orbit-specific computation is made. Because of Theorem 6.2, no matrix with $k > 5$ occurs. We refine these rank strata simultaneously by the flattening strata of the finite modules

$$\mathcal{M}_j = P_c/((d) + K_j), \quad 1 \leq j \leq 5,$$

by the flattening strata of the cokernels of (6.4), and by the associated-point stratification in the following proposition.

Proposition 6.7 (Finite intrinsic primary-signature refinement). *On every finite-type coefficient chart for (A_H, B_H, C_H) , the presentation atlas admits a finite locally closed refinement such that the following statements hold for every geometric fibre.*

- (1) *Formation of each residual colon $K_j = (J : d^{j-1})$, $1 \leq j \leq 5$, commutes with base change, and*

$$\mathcal{M}_j = P_c / ((d) + K_j)$$

is flat over the stratum with constant Hilbert polynomial.

- (2) *There are finitely many relative closed supports $Z_{j,\nu}$ whose nonempty fibrewise generic points are exactly $\text{Ass}((\mathcal{M}_j)_s)$. Their dimensions, minimal-versus-embedded status, incidence relations, and generic lengths are constant on the stratum.*
- (3) *The cokernels of the graded multiplication maps are flat, so their ranks are constant and commute with base change.*

Consequently the radicals, associated supports, generic multiplicities, nilpotency index, Hilbert polynomials, and multiplication ranks of all positive graded layers have constant intrinsic primary signature on each final stratum.

The refinement is not asserted to be canonical as a collection of coordinate strata. The primary signature is intrinsic; different Schur charts, generating sets, or monomial orders admit a common finite refinement and compute the same signature.

This refinement is used below for rank drops. The proof in Section 11 is the application of Theorem 6.6: it flattens both P_S/J and $P_S/(J, d^{j-1})$, deduces flatness of the image of multiplication by d^{j-1} , and only then invokes relative assassins. Thus coefficient collisions are recorded by actual specialization of the intrinsic graded algebra, not by relabelling a primary decomposition.

There is also a useful degree bound that detects many strata before any Macaulay matrix is formed. Put

$$a = \text{rank } A_H, \quad q = 6 - a,$$

and, after quotienting by $\text{im } A_H$,

$$b = \text{rank } \overline{B}_H.$$

Every nonzero residual q -minor uses at most b linear mixed columns and therefore has transverse degree at least

$$\ell(a, b) = 2q - b. \tag{6.7}$$

Proposition 6.8 (Rank-drop depth bounds at corank two). *At projection corank two,*

$$\min\{k : d^k \in J\} \geq \kappa_0(a, b) := q - \left\lfloor \frac{b}{2} \right\rfloor. \tag{6.8}$$

Together with $d^5 \in J$, this leaves only the following finite possibilities when A_H drops rank:

a	q	possible $(b; k)$
2	4	$(4; 2, 3, 4, 5), (3; 2; 3, 4, 5), (1; 4, 5),$
1	5	$(4; 3, 4, 5), (3; 2; 4, 5),$
0	6	$(4; 4, 5), (3; 5).$

Here $k = \min\{j : d^j \in J\}$, so the nilpotency index is $k + 1$. The omitted values of b cannot occur under surjectivity of the original quadratic quotient, because the quadratic block contributes at most three independent target directions. Each comma-separated choice of k is cut out exactly by $\varepsilon_k(J) = 0$ and $\varepsilon_{k-1}(J) = 1$. In particular the extreme rank-drop type $(a, b) = (0, 3)$ has exact nilpotency index six.

At projection corank three a stronger representation-theoretic exclusion reduces the atlas to one binary choice.

Theorem 6.9 (Projection corank three). *At every projection-corank-three point of $\hat{D}_R$, including all rank drops of A_H and of the mixed block,*

$$d^3 \notin J, \quad d^5 \in J. \quad (6.9)$$

Hence the nilpotency index is either five or six. It is five exactly when

$$\varepsilon_4(J) = 0,$$

and six exactly when $\varepsilon_4(J) = 1$. Thus the corank-three locus has two intrinsic determinant-membership strata. Any Macaulay presentation atlas computes these strata, but the chosen coefficient matrices are not themselves declared canonical.

At the deepest Schubert stratum the remaining condition is identified geometrically.

Theorem 6.10 (Projection corank four and the Jacobian quartic). *Let $R \in G_4^\circ$. At a projection-corank-four point, $H = 0$, $T \in \text{Mat}_{4 \times 4}$, and*

$$J = I_6(\gamma_R \text{Sym}^2 T).$$

Then

$$d^3 \notin J, \quad d^4 \in J, \quad d = \det T. \quad (6.10)$$

Consequently the local nilradical has exact index five.

More precisely, as a $GL(V)$ -module,

$$\bigwedge^6 \text{Sym}^2 V \simeq \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (6.11)$$

The absence of $(\det V)^3$ in (6.11) gives the first nonmembership in (6.10). Under the canonical duality

$$\bigwedge^6 \text{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \text{Sym}^2 V \right)^*,$$

the second summand in (6.11) is dual, up to determinant twist, to

$$\det V \otimes \text{Sym}^4 V,$$

and the corresponding covariant of $\bigwedge^4 R$ is the Jacobian quartic F_R . Since $F_R \neq 0$ on G_4° , the determinant line $(\det V)^4$ occurs in the degree-sixteen part of J , which is exactly the second membership in (6.10).

Theorems 6.2, 6.4, 6.9, and 6.10, together with the generic primary signature and specialization theorems of Section 11, form the boundary classification used in this revision. The residual-colon formulas are coordinate-free identities; the presentation matrices are finite computational devices. At corank two the generic supports and their primary corrections are geometrically identified, including pure-block rank drops. At higher corank the exact depth statements remain intrinsic and are coupled to the finite primary-signature refinement.

7. FIXED TENSORS ACROSS PROJECTION CORANK TWO

We now work with $e = 4$ and $p = 6$. All calculations below concern a fixed quotient tensor γ_R , while its generating plane varies. They are identities of ideals, before taking radicals.

Definition 7.1. A pair (R, H) , where $R \in G_4^\circ$ and $H \subset V$ is a plane, is called admissible for the corank-two calculation if the following conditions hold. The map $A_H = \gamma_R|_{\text{Sym}^2 H}$ is injective. For a complement W to H , put $Q_H = S_R / \text{im } A_H$ and let

$$B_H : H \otimes W \longrightarrow Q_H, \quad C_H : \text{Sym}^2 W \longrightarrow Q_H$$

be induced by multiplication. The map B_H is surjective and its one-dimensional kernel is generated by a rank-two tensor in $H \otimes W$. Finally, the line of hyperplanes containing H meets Y_R in four distinct points.

The first-order map B_H is intrinsically a map from $H \otimes (V/H)$: changing a lift of a vector in V/H adds an element of $\text{Sym}^2 H$, which is zero in Q_H . Thus these conditions are independent of the chosen complement. The last condition is transverse intersection with a projective line, not smoothness of a parameter-selected polynomial without a geometric interpretation.

Lemma 7.2 (Nonemptiness of the corank-two open). *The admissible pairs form a nonempty open subset of $G_4^\circ \times \text{Gr}(2, V)$. Its projection is a nonempty invariant open $G_4^\dagger$. For each R in that image, the admissible H form a dense open subset of $\text{Gr}(2, V)$.*

Proof. Injectivity and surjectivity are open rank conditions. After taking the line kernel, tensor rank two is the nonvanishing of its determinant. On this locus the pencil restriction of the determinant is a binary quartic; nonzero discriminant is open. These conditions are invariant. Projection from a product over $\mathbb{C}$ is open, and $\text{Gr}(2, V)$ is irreducible, giving the density claim once nonemptiness is proved.

For C_* in (13.2), take $H = \langle x_0, x_1 \rangle$ and $W = \langle x_2, x_3 \rangle$. In the square-first, cross-second basis, $\gamma = (-C_* \mid I_6)$. The three columns for x_0^2, x_0x_1, x_1^2 have rank three. After quotienting their image, the remaining maps are

$$B_H = \begin{pmatrix} -1/2 & -1 & 1 & 0 \\ -1/2 & -1 & 0 & 1 \\ -3/2 & -2 & 0 & 0 \end{pmatrix}, \quad C_H = \begin{pmatrix} -9/2 & 0 & -1/2 \\ -5/2 & 0 & 1/2 \\ -13/2 & 1 & 7/2 \end{pmatrix}. \quad (7.1)$$

The four columns of B_H correspond to $x_0x_2, x_0x_3, x_1x_2, x_1x_3$; the three of C_H correspond to x_2^2, x_2x_3, x_3^2 . The kernel vector is $(-4, 3, 1, 1)$, whose 2×2 determinant is -7 . For $ux_2 + vx_3$, the final square column is $C_H(u^2, 2uv, v^2)^t$. The determinant of that column and the two mixed columns is

$$h(u, v) = \frac{u^4 - 8u^3v - u^2v^2 - 4uv^3 - 9v^4}{2}, \quad \text{disc } h(u, 1) = -\frac{632301}{4} \neq 0. \quad (7.2)$$

Its leading coefficient is nonzero, so there is no root at infinity. This determinant is the restriction determinant on $\text{Sym}^2(H + \mathbb{C}(ux_2 + vx_3))$ after eliminating A_H , hence defines the pencil intersection with Y_R . Lemma 13.1 places this witness in G_4° . $\square$

Lemma 7.3 (The residual matrix). *Near a plane K whose projection has admissible rank-two image H , there are smooth auxiliary parameters and four transverse coordinates a, b, c, d in which $\mathcal{I}_{\widehat{D}_R} = \delta J$, with*

$$\begin{aligned} T &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d), \\ L(T) &= \begin{pmatrix} a & b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & c & d \end{pmatrix}, \quad T^{(2)} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad + bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}, \\ J &= I_3[L(T), CT^{(2)}]. \end{aligned} \quad (7.3)$$

Here C is a 3×3 matrix depending only on the auxiliary parameters, not on a, b, c, d .

Proof. Use Proposition 3.1 and an invertible projection two-minor to write $M = \text{diag}(I_2, T)$. The basis of its first image plane and a complement can be chosen independently of the Schur block T : the Gaussian row operations depend on the first two columns, and the remaining column operations remove their components from the last two columns. This proves the asserted separation of parameters.

The quadratic columns divide into $\text{Sym}^2 H$, $H \otimes W$, and $\text{Sym}^2 W$. Eliminate the injective rank-three A_H block. The residual matrix is $[B_H(I_H \otimes T), C_H T^{(2)}]$, and its maximal minors

generate the residual ideal: the eliminated identity block removes a free direct summand of the cokernel. A rank-two generator of $\ker B_H$ can be put in the form $h_1 \otimes w_2 - h_2 \otimes w_1$ by invertible changes of bases. The quotient is then the symmetrization map from two two-dimensional spaces, with the displayed matrix $L(T)$. These basis changes involve only the auxiliary parameters. Finally $\det M$ is a unit times δ . This proves the ideal identity, not just its leading term. $\square$

Let

$$h(u_1, u_2) = \det \left[\begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ u_1 \\ u_2 \end{pmatrix}, C \begin{pmatrix} u_1^2 \\ 2u_1u_2 \\ u_2^2 \end{pmatrix} \right]. \quad (7.4)$$

It is the binary quartic of the pencil intersection. After an étale extension of the auxiliary coefficient ring, its four roots split as distinct line subbundles $\ell_i \subset W$. Write P_i for the ideal of $\text{im } T \subset \ell_i$; it is generated by the two entries of the composite $\mathbb{C}^2 \xrightarrow{T} W \rightarrow W/\ell_i$.

Lemma 7.4 (A rank-one-cone computation). *Under the simple-root hypothesis, in the local coefficient ring of Lemma 7.3 one has*

$$I := \bigcap_{i=1}^4 P_i = (\delta, g_0, \dots, g_4), \quad J = I \cap \mathfrak{m}^3 = \delta \mathfrak{m} + (g_0, \dots, g_4). \quad (7.5)$$

The degree-four polynomials g_j can be chosen with restrictions $h(u_1, u_2)v_1^{4-j}v_2^j$ on $T = uv^t$.

Proof. Give a, b, c, d degree one and the auxiliary coefficients degree zero. The coordinate ring of the rank-one cone is the diagonal subring

$$A_0[a, b, c, d]/(\delta) \simeq \bigoplus_{r \geq 0} \text{Sym}^r(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v, \quad T = uv^t, \quad (7.6)$$

where A_0 denotes the auxiliary ring. This follows directly from $a = u_1v_1$, $b = u_1v_2$, $c = u_2v_1$, $d = u_2v_2$ and the single relation $ad - bc$; the monomials on either side give the same basis. In degree r , vanishing on the four distinct u -lines means divisibility by h in the u variables. The resulting module is zero for $r < 4$, and for $r \geq 4$ it is

$$h \text{Sym}^{r-4}(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v.$$

The simple-root discriminant is a unit, so the elementary divisibility and interpolation assertions used here hold over A_0 as well as over its residue field. Multiplication of binary forms shows that this ideal is generated in diagonal degree four by $hv_1^{4-j}v_2^j$. Lifting those generators proves the first identity.

The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, by direct expansion. Modulo δ , its four columns are

$$v_1B(h_1u), \quad v_2B(h_1u), \quad v_1B(h_2u), \quad v_2B(h_2u),$$

and the three quadratic columns are $v_1^2C(u^2)$, $v_1v_2C(u^2)$, and $v_2^2C(u^2)$. Here u^2 has coefficient vector $(u_1^2, 2u_1u_2, u_2^2)^t$. A minor using one column from each of the two linear groups and one quadratic column is h times a degree-four monomial in v ; these span all five such monomials. Thus they furnish the lifts g_j . A minor with at least two quadratic columns vanishes modulo δ , and its degree is at least five, so it already lies in $\delta \mathfrak{m}$. All other minors have been accounted for. This proves $J = \delta \mathfrak{m} + (g_j)$ exactly.

Finally I is generated by its degree-two element δ and the g_j of degree four. Its intersection with $\mathfrak{m}^3$ deletes only the scalar multiple of δ in degree two, proving the second equality. Localizing in the auxiliary parameters or taking the indicated étale extension preserves these explicit identities. $\square$

Theorem 7.5 (The corank-two primary structure). *For an admissible pair the ideal δJ has the decomposition (1.4). It is irredundant. In the local failure ring its nilradical $\widehat{\mathcal{N}}$ satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J). \quad (7.7)$$

Consequently the rank-two stratum is intrinsic to this local scheme as the annihilator support of $\widehat{\mathcal{N}}^2$.

Proof. Work on a connected regular étale auxiliary chart, so that its coefficient ring is a domain. Each P_i is a prime generated by two independent linear coordinates. Its ordinary square is P_i -primary; the same argument applies to $\mathfrak{m}^5$. The ideal (δ) is prime. At the generic point of P_i , the determinant has order one, because the other row after choosing ℓ_i as a coordinate line is not zero. Therefore

$$(P_i^2 : \delta) = P_i, \quad (\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

The second equality also follows immediately from the homogeneous order two of δ and the polynomial associated graded ring of $\mathfrak{m}$. Intersecting with (δ) and cancelling δ now gives

$$(\delta) \cap \bigcap_i P_i^2 \cap \mathfrak{m}^5 = \delta \left(\bigcap_i P_i \cap \mathfrak{m}^3 \right) = \delta J$$

by Lemma 7.4.

At the generic point of the determinant divisor all the other ideals are units, so (δ) is necessary. At the generic point of P_i the other P_j and $\mathfrak{m}$ become units, and the local ideal is $\delta P_i = (\delta) \cap P_i^2$; its two components are necessary. To see that $\mathfrak{m}^5$ is necessary, note that δ^2 lies in (δ) and in every P_i^2 , but is not in δJ : $\delta \notin J$ because its degree is two. On the other hand $\mathfrak{m}\delta^2 \subset \delta J$ since $\delta\mathfrak{m} \subset J$. This also proves that the annihilator of the class of δ^2 is exactly $\mathfrak{m}$; a nonzero auxiliary coefficient cannot kill that degree-four class.

The radical of the failure ideal is (δ) , so the nilradical is generated by the class of δ . Its square is the nonzero class just considered and its cube vanishes, since $\delta^2 \in \delta\mathfrak{m} \subset J$. This proves (7.7). Over the auxiliary ring the square is free of rank one on the rank-two stratum. Intrinsically its transition functions are those of $\mathcal{I}_{\Delta}^2$ restricted to that stratum; thus it is $\mathcal{O}_{\Sigma}(-2\widehat{\Delta})$ on the admissible open Σ . The generic transverse length is one. $\square$

Proof of Theorem 1.3. Lemma 7.2 supplies the asserted nonempty open class. Lemmas 7.3 and 7.4 and Theorem 7.5 supply the fixed-tensor calculation on every indicated frame chart. The graph coordinates are smooth auxiliary variables, so no transverse component has been discarded. The individual P_i are allowed to permute under monodromy; the intersection and the intrinsic annihilator in (7.7) descend. Four local branches need not be four distinct global irreducible components. This is why the primary decomposition was stated étale locally rather than with globally labelled roots. $\square$

8. RESIDUAL PRESENTATIONS AND THE RANK-TWO CONTACT BOUNDARY

We first record the projection-rank block presentation inherited from the full Fitting formula, and then isolate the rank-two structure to which it gives rise. The distinction between a presentation theorem and a structure theorem is important: the block formula below is exact at every projection rank, whereas the associated-prime, collision, and nilpotent-filtration theorems are proved on the loci explicitly stated. Section 9 pushes the structure theorem through all mixed-regular contact collisions, through the containment locus, and across the first non-mixed-regular tensor-rank wall.

For $0 \leq r \leq 3$, let $\Sigma_R^{(r)} \subset X_R$ be the locally closed stratum on which the projection $\mathcal{K} \rightarrow V$ has rank exactly r . If $K \in \Sigma_R^{(r)}$ has image $H \subset V$, choose a complement $V = H \oplus W$, where

$c = \dim W = 4 - r$. On a frame chart around K , an invertible r -minor and Schur reduction put the projection block in the form

$$M = \begin{pmatrix} I_H & 0 \\ 0 & T \end{pmatrix}, \quad T \in \text{Mat}_{c \times c}.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write the corresponding blocks as A_H, B_H, C_H .

Theorem 8.1 (Projection-rank residual block presentation). *For every $R \in G_4^2$, every $0 \leq r \leq 3$, and every $K \in \Sigma_R^{(r)}$, one has on the preceding frame chart, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\det T) I_6 [A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T]. \quad (8.1)$$

The formula is independent of any injectivity, mixed-rank, tensor-rank, or contact-transversality assumption. Thus every projection-rank stratum has the same exact block presentation. No claim of a primary or associated-prime classification on the non-mixed-regular corank-three or corank-four boundary is made by this formula alone; the separate structure theorems below derive additional information where the residual algebra can be determined.

Proof. Proposition 3.1 gives on every frame chart

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6 (\gamma_R \text{Sym}^2 M).$$

The Schur operations placing M in the displayed block form are invertible and therefore change neither the maximal-minor ideal nor the local scheme. On $\text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$ the induced map $\text{Sym}^2 M$ is respectively

$$I_{\text{Sym}^2 H}, \quad 1_H \otimes T, \quad \text{Sym}^2 T.$$

Since $\det M$ is a unit times $\det T$, substitution proves (8.1). $\square$

We now specialize this exact all-corank formula to $r = 2$. Write $\Sigma_R = \Sigma_R^{(2)}$. Then $c = 2$, and after choosing bases

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = \det T = ad - bc.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write

$$A_H : \text{Sym}^2 H \rightarrow S_R, \quad B_H : H \otimes W \rightarrow S_R, \quad C_H : \text{Sym}^2 W \rightarrow S_R.$$

Theorem 8.2 (Exact presentation on every rank-two chart). *In the preceding coordinates one has, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) I_6 [A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T]. \quad (8.2)$$

The formula is valid at every projection-rank-two point; no injectivity, mixed-rank, tensor-rank, or simple-contact hypothesis is required.

Proof. Proposition 3.1 gives

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6 (\gamma_R \text{Sym}^2 M).$$

For $M = I_H \oplus T$, the induced map on the three summands of $\text{Sym}^2(H \oplus W)$ is respectively

$$I_{\text{Sym}^2 H}, \quad 1_H \otimes T, \quad \text{Sym}^2 T.$$

Substitution gives (8.2). All row and column operations used to reach the block form are invertible, so the maximal-minor ideal is unchanged. $\square$

Thus the exceptional rank-two locus is included in the exact presentation. Determining radicals, associated primes, and nilpotent layers of its residual factor is a separate problem. The mixed-regular case and its first tensor-rank boundary are solved below and in Section 9.

Definition 8.3. A pair (R, H) is *mixed-regular* if A_H is injective and, after quotienting S_R by its image, the induced map

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \text{im } A_H$$

is surjective with one-dimensional kernel generated by a rank-two tensor. No hypothesis is imposed on the multiplicities of the four ramification contacts.

These are precisely the first three conditions in Definition 7.1; the discriminant condition has been removed. On this locus the same Gaussian elimination as in Lemma 7.3 is valid and yields

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = I_3[L(T), C T^{(2)}],$$

with $L(T)$ and $T^{(2)}$ as in (7.3).

Lemma 8.4 (Root-free residual normal form). *Over the local auxiliary coefficient ring of a mixed-regular pair,*

$$J = \delta \mathfrak{m} + (g_0, \dots, g_4), \quad \mathfrak{m} = (a, b, c, d), \quad (8.3)$$

where each g_i is homogeneous of transverse degree four. Modulo (δ) , under the Segre parametrization $T = uv^t$, their classes span

$$h_H(u) \cdot \text{Sym}^4(\mathbb{C}_v^2).$$

Here h_H is the binary quartic obtained by restricting the ramification equation to the pencil of hyperplanes containing H . The identity is valid with repeated roots, and also when the discriminant of h_H vanishes.

Proof. The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, exactly as in Lemma 7.4. Modulo δ , write $T = uv^t$. The two pairs of linear columns are scalar multiples, by v_1, v_2 , of the two mixed columns determined by u , while the three quadratic columns are scalar multiples, by $v_1^2, v_1 v_2, v_2^2$, of $C(u^2)$. A minor containing one column from each linear pair and one quadratic column is therefore $h_H(u)$ times a degree-four monomial in v ; the five choices span $\text{Sym}^4(\mathbb{C}_v^2)$.

Every minor containing at least two quadratic columns vanishes on the rank-one cone, hence is divisible by the prime equation δ . Its transverse degree is at least five, so the quotient by δ lies in $\mathfrak{m}$. The remaining minors are the already listed linear and mixed ones. This proves (8.3). No factorization of h_H was used. When its roots are distinct, factorizing h_H recovers Lemma 7.4. $\square$

Theorem 8.5 (Depth two across contact collisions). *At every mixed-regular projection-rank-two point, the radical of $\mathcal{I}_{\widehat{D}_R}$ is (δ) and the nilradical $\widehat{\mathcal{N}}$ of the local failure ring satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J).$$

Consequently $\mathcal{Z}_2(\widehat{D}_R)$ is the rank-two stratum on the mixed-regular locus, even when the contact quartic has multiple roots. Moreover

$$\widehat{\mathcal{N}}^2|_{\Sigma_R} \simeq \mathcal{I}_{\Delta_R}^2|_{\Sigma_R}.$$

Proof. By Lemma 8.4, $J \subset \mathfrak{m}^3$ and $\delta \mathfrak{m} \subset J$. Since $\delta \in \mathfrak{m}^2$, we have $\delta^2 \in J$, hence $\delta^3 \in \delta J$. On the other hand $\delta \notin J$ by transverse degree, so $\delta^2 \notin \delta J$. Since $\delta J \subset (\delta)$ and a power of δ belongs to δJ , the radical is exactly (δ) . Thus the nilradical is generated by the class of δ and has exact index three.

For the annihilator, cancellation in the regular local coefficient ring gives

$$(\delta J : \delta^2) = (J : \delta).$$

The inclusion $\mathfrak{m} \subset (J : \delta)$ follows from $\delta \mathfrak{m} \subset J$. Conversely, use the transverse grading: J has no component of transverse degree two, whereas δ has degree two. If the transverse degree-zero part of r were nonzero, the degree-two part of $r\delta$ could not lie in J . Hence $r \in \mathfrak{m}$, proving $(J : \delta) = \mathfrak{m}$.

Finally δ is a local equation of the Schubert divisor $\widehat{\Delta}_R$. Therefore the generator δ^2 of the square of the nilradical transforms as the square of the conormal line. This gives the displayed line bundle after restriction to the rank-two stratum. $\square$

We next globalize the four contacts without labelling them. Put

$$G = \mathrm{Gr}(2, V)$$

and let $\mathcal{Q}$ be its universal rank-two quotient bundle. The quartic equation F_R of $Y_R \subset \mathbb{P}(V^*)$ restricts to every line of hyperplanes containing H . Equivalently it defines a global binary quartic

$$h_R \in H^0(G, \mathrm{Sym}^4 \mathcal{Q}). \quad (8.4)$$

Define the contact incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}. \quad (8.5)$$

Proposition 8.6 (The global contact cover). *The projection $\mathcal{R}_R \rightarrow Y_R$ is a $\mathbb{P}^2$ -bundle; in particular $\mathcal{R}_R$ is smooth and irreducible. Over the open set $G_R^{\mathrm{fin}} \subset G$ on which the binary quartic (8.4) is not identically zero, the map*

$$\mathcal{R}_R \longrightarrow G$$

is finite flat of degree four. There is a globally defined discriminant divisor

$$\mathfrak{D}_R = V(\mathrm{Disc}(h_R)), \quad \mathrm{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}).$$

Whenever this section is not identically zero, $[\mathfrak{D}_R] = 12H$ in the Pluecker class group. On the finite-flat locus G_R^{fin} , the support of the branch divisor is the restriction of $\mathfrak{D}_R$; away from $\mathfrak{D}_R$ the degree-four cover is finite etale. The monodromy action on the four sheets over that etale locus is transitive.

Proof. For $[\alpha] \in Y_R$, the fibre of $\mathcal{R}_R \rightarrow Y_R$ consists of the two-planes in the three-dimensional hyperplane $\ker \alpha$, hence is $\mathrm{Gr}(2, \ker \alpha) \simeq \mathbb{P}^2$. These fibres form the relative Grassmannian of the rank-three hyperplane bundle on Y_R , proving smoothness and irreducibility.

For fixed H , the other projection has fibre $Y_R \cap \mathbb{P}(H^\perp)$, the zero scheme of the restricted quartic. On the locus where the restriction is nonzero this is a relative effective Cartier divisor of degree four in the $\mathbb{P}^1$ -bundle $\mathbb{P}(\mathcal{Q}^*)$, hence finite flat of degree four.

The discriminant of a binary quartic is homogeneous of degree six in its coefficients and is a relative invariant of determinant weight twelve. Applied to the section of $\mathrm{Sym}^4 \mathcal{Q}$ in (8.4), it is therefore a section of $(\det \mathcal{Q})^{12}$. Since $\det \mathcal{Q}$ is the Pluecker line bundle up to the constant factor $\det V$, its zero divisor has class $12H$ whenever it is nonzero. On G_R^{fin} the fibre is a degree-four Cartier divisor on a projective line, so its non-etale locus is exactly the multiple-root locus, i.e. the restriction of the discriminant locus. The total incidence is irreducible; therefore the monodromy action of the etale restriction on a geometric fibre is transitive. $\square$

For R in the nonempty open $G_4^\dagger$ of Lemma 7.2, the discriminant in Proposition 8.6 is not identically zero. Let $G_R^{\mathrm{mr}} \subset G$ denote the mixed-regular locus of Definition 8.3. The contact sheets themselves split etale locally over $G_R^{\mathrm{fin}} \setminus \mathfrak{D}_R$. Their interpretation as the four primary ideals

P_i of Theorem 7.5, however, also requires the independent mixed-regular rank conditions. Thus the exact locus for that primary decomposition is

$$G_R^{\text{mr}} \cap (G_R^{\text{fin}} \setminus \mathfrak{D}_R),$$

not the discriminant complement by itself. On this intersection the P_i are the four etale contact sheets and their permutation is the monodromy of the canonical cover (8.5).

The collision divisor has the five nonzero fibre types corresponding to the partitions

$$1 + 1 + 1 + 1, \quad 2 + 1 + 1, \quad 2 + 2, \quad 3 + 1, \quad 4$$

of the degree of a binary quartic, together with the separate containment locus $h_H \equiv 0$. Theorem 8.5 does not change across these collision types. What changes is the primary splitting of the contact part, so we do not force the simple-root intersection formula onto the boundary.

The ordinary fifth power of the rank-two Schubert ideal is of course globally defined. The stronger assertion that it remains the actual $\mathfrak{m}$ -primary component of the failure ideal through every mixed-regular collision, including the zero-quartic containment locus, is proved in Theorem 9.2; it is not inferred merely from the existence of the global ideal. The same theorem gives the exact collided primary types and separates the finite contact geometry from the containment geometry.

Corollary 8.7 (Rank-two part of Theorem 1.2). *The contact-incidence and collision assertions of Theorem 1.2 hold on the stated mixed-regular locus.*

Proof. Theorem 8.5 identifies depth two and its line bundle, while Proposition 8.6 globalizes the contact incidence and discriminant. The global residual-colon filtration, the simultaneous identification of the first two graded pieces, the collided primary components, the containment locus, and the decomposable-kernel wall are proved in Section 9. $\square$

9. THE STRATIFIED NILPOTENT ALGEBRA AT PROJECTION CORANK TWO

The root-free formula of Lemma 8.4 contains more scheme structure than the nilpotency-index statement extracted from it in the preceding revision. We record that structure before passing to the exceptional mixed-kernel boundary.

Theorem 9.1 (Global residual-colon filtration). *Let $X = X_R$, let $\hat{\Delta}_R \subset X$ be the reduced Schubert divisor, and put*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\hat{\Delta}_R} : \mathcal{I}_{\hat{\Delta}_R}) \subset \mathcal{O}_X.$$

Then

$$\mathcal{I}_{\hat{\Delta}_R} = \mathcal{I}_{\hat{\Delta}_R} \mathcal{J}_R^{\text{res}}. \quad (9.1)$$

For every $j \geq 1$,

$$\text{Ann}_{\mathcal{O}_{\hat{\Delta}_R}}(\hat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\hat{\Delta}_R}^{j-1})}{\mathcal{I}_{\hat{\Delta}_R}}. \quad (9.2)$$

Define the closed subscheme of the reduced divisor

$$W_j = V_{\hat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\hat{\Delta}_R}^{j-1}) + \mathcal{I}_{\hat{\Delta}_R} \right). \quad (9.3)$$

If

$$\mathcal{L} = \mathcal{I}_{\hat{\Delta}_R} / \mathcal{I}_{\hat{\Delta}_R}^2$$

is the conormal line of the Schubert divisor, then there are canonical isomorphisms

$$\hat{\mathcal{N}}_R^j / \hat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes_{\mathcal{O}_{\hat{\Delta}_R}} \mathcal{O}_{W_j}. \quad (9.4)$$

Consequently

$$\mathrm{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (9.5)$$

with the multiplication induced by multiplication of powers of $\mathcal{I}_{\widehat{\Delta}_R}$. The sum is finite.

These formulas hold on the whole Grassmannian, including every projection corank and every rank-drop locus. On a Schur chart with local equation $d = \det T$ and residual ideal J , they become

$$\mathrm{Ann}(N^j) = (J : d^{j-1})/(dJ), \quad N^j/N^{j+1} \simeq A/((d) + (J : d^{j-1})). \quad (9.6)$$

In particular the nilpotency index at the point is

$$1 + \min\{k \geq 0 : d^k \in J\}. \quad (9.7)$$

Proof. Proposition 3.1 shows locally that $\mathcal{I}_{\widehat{D}_R} = dJ$, where d is a local equation of the Cartier divisor $\widehat{\Delta}_R$. Since d is a non-zero-divisor in the regular local ring of X ,

$$(dJ : d) = J.$$

Thus J is the local expression of the intrinsic colon $\mathcal{J}_R^{\mathrm{res}}$, and the local identities glue to (9.1).

In $A/(dJ)$ the nilradical is the image of (d) , because the reduction is $A/(d)$. Therefore

$$\mathrm{Ann}(N^j) = (dJ : d^j)/(dJ) = (J : d^{j-1})/(dJ),$$

which proves (9.2). For the graded piece, the map

$$A \longrightarrow N^j/N^{j+1}, \quad r \longmapsto rd^j$$

has kernel

$$(d) + (J : d^{j-1}).$$

Indeed

$$rd^j \in d^{j+1} + dJ \iff rd^{j-1} \in d^j + J,$$

and the latter is equivalent to writing $r = ds + t$ with $td^{j-1} \in J$. This proves the local form of (9.4). On overlaps local equations of the Cartier divisor differ by units, and d^j transforms by the j -th tensor power of the conormal transition function. Hence the local identifications glue to (9.4) and (9.5). Nilpotence of the nilradical makes the sum finite, and the same cancellation gives (9.7). $\square$

The theorem converts the globally defined annihilator filtration into an exact residual-colon filtration. What remains is to identify the colon schemes W_j geometrically on the successive rank strata. For projection corank one, $W_1 = E_R$ and all higher W_j are empty. The next two results compute W_1, W_2 through the entire mixed-regular rank-two boundary and then across its first tensor-rank wall.

Work on a mixed-regular rank-two chart. Let A_0 be the regular auxiliary coefficient ring, put

$$P = A_0[a, b, c, d], \quad \mathfrak{m} = (a, b, c, d), \quad \delta = ad - bc,$$

and localize, or complete, at the point under consideration. Thus

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = \delta \mathfrak{m} + (g_0, \dots, g_4),$$

where the g_i have transverse degree four. Define the *unsplit contact ideal*

$$I_H = (\delta, g_0, \dots, g_4). \quad (9.8)$$

Modulo δ , the Segre parametrization $T = uv^t$ identifies the homogeneous coordinate ring with

$$P/(\delta) \simeq \bigoplus_{n \geq 0} \mathrm{Sym}^n(\mathbb{C}_u^2) \otimes \mathrm{Sym}^n(\mathbb{C}_v^2),$$

and the image of I_H in degree n is zero for $n < 4$ and

$$h_H \operatorname{Sym}^{n-4}(\mathbb{C}_u^2) \otimes \operatorname{Sym}^n(\mathbb{C}_v^2) \quad (n \geq 4). \quad (9.9)$$

Theorem 9.2 (Saturated contact ideal and the fixed vertex component). *At every mixed-regular rank-two point one has the exact identities*

$$J = I_H \cap \mathfrak{m}^3 \quad (9.10)$$

and

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (9.11)$$

If $h_H \not\equiv 0$, the ideal I_H is $\mathfrak{m}$ -saturated. Put $\mathcal{C}_H = V(I_H)$ for the affine contact cone. Its projectivization on the Segre quadric

$$Q = \mathbb{P}_u^1 \times \mathbb{P}_v^1 \subset \mathbb{P}^3$$

is the effective Cartier divisor

$$C_H^{\text{proj}} = V(h_H) \times \mathbb{P}_v^1 \quad (9.12)$$

of class $(4, 0)$. Consequently δI_H is also $\mathfrak{m}$ -saturated, and $\mathfrak{m}^5$ in (9.11) is an actual irredundant $\mathfrak{m}$ -primary component, not merely a globally defined ideal.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_{\nu} e_\nu = 4,$$

over an etale coefficient extension, and P_ν denotes the height-two contact prime corresponding to ℓ_ν , then

$$\operatorname{Ass}(P/\mathcal{I}_{\widehat{D}_R}) = \{(\delta), P_1, \dots, P_s, \mathfrak{m}\}. \quad (9.13)$$

At the generic point of P_ν , choose a second regular parameter q_ν so that $P_\nu = (\delta, q_\nu)$. The P_ν -primary part of δI_H has the exact two-parameter normal form

$$Q_{\nu, e_\nu} = (\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (9.14)$$

For $e_\nu = 1$ this is P_ν^2 ; for a collided contact it is the corresponding non-ordinary primary thickening.

If $h_H \equiv 0$, then

$$I_H = (\delta), \quad J = \delta \mathfrak{m}, \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (9.15)$$

In particular the rank-two vertex component remains $\mathfrak{m}^5$ on the containment locus.

Proof. The equality (9.10) is a graded calculation. The ideal I_H is generated in transverse degrees two and four, its only degree-two generator being δ . Intersecting with $\mathfrak{m}^3$ therefore removes exactly the scalar multiple of δ and leaves $\delta \mathfrak{m} + (g_0, \dots, g_4) = J$.

Since the initial form of δ has degree two and is nonzero in the polynomial associated graded ring,

$$(\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

Cancellation in the regular ring gives

$$\delta I_H \cap \mathfrak{m}^5 = \delta(I_H \cap (\mathfrak{m}^5 : \delta)) = \delta(I_H \cap \mathfrak{m}^3) = \delta J,$$

which proves (9.11).

Assume first $h_H \neq 0$. Formula (9.9) is precisely the section ideal of $\mathcal{O}_Q(-4, 0) \subset \mathcal{O}_Q$, cut out by the nonzero section h_H of $\mathcal{O}_Q(4, 0)$. Hence it is saturated and defines the Cartier divisor (9.12). The inverse image $I_H \subset P$ is therefore $\mathfrak{m}$ -saturated. The product δI_H is saturated as well: if $\mathfrak{m}^N f \subset \delta I_H$, choose a coordinate not divisible by the prime element δ to see that $\delta \mid f$; after

cancelling δ , saturation of I_H applies. Thus $\mathfrak{m} \notin \text{Ass}(P/\delta I_H)$. On the other hand $\delta^2 \in \delta I_H$ has transverse degree four and does not belong to $\mathfrak{m}^5$, so the $\mathfrak{m}^5$ factor in (9.11) is redundant.

The divisor C_H^{proj} has components $e_\nu(\{\ell_\nu = 0\} \times \mathbb{P}^1)$. Hence P/I_H is Cohen–Macaulay of pure codimension two and its associated primes are exactly the P_ν . Multiplication by δ gives an exact sequence

$$0 \longrightarrow P/I_H \xrightarrow{\delta} P/\delta I_H \longrightarrow P/(\delta) \longrightarrow 0.$$

It follows, and localization shows equality, that the associated primes of $P/\delta I_H$ are $(\delta), P_1, \dots, P_s$. Intersecting with the irredundant $\mathfrak{m}$ -primary ideal $\mathfrak{m}^5$ gives (9.13).

At the generic point of P_ν , the Cartier divisor has equation $q_\nu^{e_\nu}$, so

$$(I_H)_{P_\nu} = (\delta, q_\nu^{e_\nu}), \quad (\delta I_H)_{P_\nu} = (\delta^2, \delta q_\nu^{e_\nu}).$$

The ideal

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1})$$

is P_ν -primary, and its colon by δ is $(\delta, q_\nu^{e_\nu})$. Therefore

$$(\delta) \cap Q_{\nu, e_\nu} = \delta(\delta, q_\nu^{e_\nu}),$$

which proves the asserted generic primary normal form.

If $h_H = 0$, every g_i vanishes modulo δ . Since it has transverse degree four, $g_i = \delta q_i$ with $q_i \in \mathfrak{m}^2$. Thus $I_H = (\delta)$ and $J = \delta \mathfrak{m}$. Finally

$$(\delta^2) \cap \mathfrak{m}^5 = \delta^2(\mathfrak{m}^5 : \delta^2) = \delta^2 \mathfrak{m},$$

again by the polynomial associated graded ring. This is (9.15). $\square$

The preceding theorem gives the requested specialization law without labelling roots. A collision changes the Cartier multiplicities in C_H and hence the primary type (9.14); it does not change the vertex component. The zero quartic is the unique point at which the Cartier contact divisor ceases to be a divisor and fills the whole Segre quadric.

Theorem 9.3 (Associated graded nilpotent algebra across coranks one and two). *On a mixed-regular rank-two chart let*

$$A = P/(\delta J), \quad N = \sqrt{0_A}.$$

Then $N = (\delta)/(\delta J)$, $N^3 = 0 \neq N^2$, and

$$N/N^2 \simeq P/I_H, \tag{9.16}$$

$$N^2 \simeq P/\mathfrak{m}. \tag{9.17}$$

Equivalently, with $\mathcal{L}$ the conormal line of the Schubert divisor, the associated graded algebra has the intrinsic form

$$\text{gr}_N \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{C_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}) \tag{9.18}$$

on the mixed-regular neighbourhood, with multiplication induced by the incidence of the affine contact cone $\mathcal{C}_H$ with the rank-two vertex Σ_R . Moreover

$$\text{Ann}_A(N) = J/(\delta J), \quad \text{Ann}_A(N^2) = \mathfrak{m}/(\delta J), \tag{9.19}$$

so

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = V(\mathfrak{m}).$$

On the punctured determinant cone, i.e. on the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is exactly the ramification support E_R . Thus $\mathcal{C}_H$ is its scheme-theoretic closure away from the vertex, while $\mathcal{Z}_1$ records in addition the canonical third-order vertex thickening forced by the Fitting ideal.

Proof. The radical of δJ is (δ) , so the nilradical is generated by the class of δ . The isomorphism (9.16) follows by cancelling one factor:

$$N/N^2 = \frac{(\delta)}{(\delta^2) + (\delta J)} \simeq \frac{P}{(\delta) + J} = \frac{P}{I_H}.$$

For the square, multiplication by δ^2 identifies its kernel with

$$(\delta J : \delta^2) = (J : \delta) = \mathfrak{m}$$

by Theorem 8.5; hence (9.17). The transition functions of the local generator δ are those of the conormal line, giving (9.18).

Cancellation also gives

$$(\delta J : \delta) = J,$$

while the preceding colon calculation gives the second formula in (9.19). Away from the vertex $\mathfrak{m}$ becomes the unit ideal, so $J = I_H$. On the smooth rank-one part of the determinant divisor, the corank-one normal form $t(t, f)$ identifies this contact equation with the ramification equation. Hence the restriction is precisely E_R . $\square$

We finally cross the first non-mixed-regular wall. This is the rank-one tensor wall in the one-dimensional mixed kernel, explicitly requested by the referee.

Definition 9.4. Assume A_H is injective and $\overline{B}_H$ is surjective with one-dimensional kernel generated by a decomposable tensor $h_1 \otimes w_1$. Choose h_2, w_2 completing bases and put

$$q_1 = \overline{B}_H(h_1 \otimes w_2), \quad q_2 = \overline{B}_H(h_2 \otimes w_1).$$

We call the point *transverse decomposable-kernel regular* if

$$\det[q_1, q_2, \overline{C}_H(w_1^2)] \neq 0.$$

This condition is independent of the chosen completing vectors. It says that the contact root forced by the decomposable kernel is simple in the direction transverse to the kernel wall.

Theorem 9.5 (The decomposable mixed-kernel wall). *At a transverse decomposable-kernel regular rank-two point there are bases for which*

$$L_{\text{dec}}(T) = \begin{pmatrix} c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}. \quad (9.20)$$

Let

$$J_{\text{dec}} = I_3[L_{\text{dec}}(T), C T^{(2)}], \quad K_{\text{dec}} = (c, d, a^2, ab, b^2).$$

Then

$$(J_{\text{dec}} : \delta) = K_{\text{dec}}. \quad (9.21)$$

Hence the local failure ring still has nilradical of exact index three, but its second intrinsic depth support is no longer reduced:

$$\mathcal{Z}_2 = V(K_{\text{dec}}). \quad (9.22)$$

Its support is the rank-two vertex, and its transverse local length is three. Thus crossing the tensor-rank-two wall changes the scheme structure of $\mathcal{Z}_2$ before it changes its set-theoretic support.

Proof. The stated normal form is obtained by taking $\ker \overline{B}_H = \mathbb{C}(h_1 \otimes w_1)$ and using

$$q_1, q_2, q_3 = \overline{B}_H(h_2 \otimes w_2)$$

as target basis. Applying $1_H \otimes T$ gives (9.20). Its maximal minors generate

$$I_3(L_{\text{dec}}) = \delta(c, d).$$

After replacing the first quadratic target column by its component modulo $\langle q_1, q_2 \rangle$, the transversality assumption allows us to normalize $\overline{C}_H(w_1^2) \equiv q_3$. The minors containing one column from each mixed group and this quadratic column give

$$\delta a^2, \quad \delta ab, \quad \delta b^2 \in J_{\text{dec}}.$$

Together with $I_3(L_{\text{dec}})$ this proves $\delta K_{\text{dec}} \subset J_{\text{dec}}$, hence $K_{\text{dec}} \subset (J_{\text{dec}} : \delta)$.

For the reverse inclusion use the transverse grading. The degree-three part of J_{dec} is exactly $\delta(c, d)$; every minor using a quadratic column has degree at least four. Modulo K_{dec} , every class is represented by

$$\lambda + \mu a + \nu b.$$

If $r\delta \in J_{\text{dec}}$, the degree-two term rules out $\lambda \neq 0$, while the degree-three identity

$$(\mu a + \nu b)\delta \in \delta(c, d)$$

forces $\mu = \nu = 0$. Thus $r \in K_{\text{dec}}$, proving (9.21).

Since $\delta \in (c, d)$, one has $\delta^2 \in \delta(c, d) \subset J_{\text{dec}}$, whereas $\delta \notin J_{\text{dec}}$ by degree. Therefore the nilradical of $P/(\delta J_{\text{dec}})$ has exact index three. Finally

$$\text{Ann}(N^2) = (\delta J_{\text{dec}} : \delta^2) = (J_{\text{dec}} : \delta) = K_{\text{dec}}.$$

The quotient by K_{dec} has basis $1, a, b$, proving the length assertion. $\square$

Proposition 9.6 (The containment locus is the finite line scheme). *For fixed $R \in G_4^\circ$, the locus*

$$\mathcal{C}_R = \{H \in \text{Gr}(2, V) : h_H \equiv 0\}$$

is canonically the Fano scheme $F_1(Y_R)$ of lines on the smooth quartic Y_R . It is finite and reduced. At its mixed-regular points the local failure ideal is (9.15), and the contact morphism $\mathcal{R}_R \rightarrow \text{Gr}(2, V)$ has fibre $\mathbb{P}^1$, rather than a finite length-four fibre.

Proof. The line of hyperplanes containing H is $\mathbb{P}(H^\perp) \subset \mathbb{P}(V^*)$. Its quartic restriction is h_H , so $h_H = 0$ identically exactly when $\mathbb{P}(H^\perp) \subset Y_R$. This identifies $\mathcal{C}_R$ with $F_1(Y_R)$.

If $L \subset Y_R$ is such a line, the normal sequence in $\mathbb{P}^3$ gives

$$\deg N_{L/Y_R} = \deg N_{L/\mathbb{P}^3} - \deg N_{Y_R/\mathbb{P}^3}|_L = 2 - 4 = -2.$$

Since N_{L/Y_R} has rank one, $N_{L/Y_R} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$, and therefore $H^0(N_{L/Y_R}) = 0$. The Zariski tangent space of the Fano scheme vanishes at every point. The Fano scheme is projective and zero-dimensional; a zero-dimensional local algebra with zero tangent space is a field by Nakayama. Hence it is finite reduced. The final two assertions follow from Theorem 9.2 and the definition of the incidence fibre. $\square$

10. PROOFS FOR THE INTRINSIC BOUNDARY ATLAS

We prove the assertions of Section 6. All calculations are made in the regular coefficient ring of a Schur chart. Since the identities are invariant under changes of frames, they descend by the same argument as the Fitting identities in Proposition 3.1.

10.1. Universal determinant completion.

Proof of Theorem 6.1. Put $N = \dim \text{Sym}^r V$ and $s = \dim S$. Choose a vector space K of dimension $N - s$ and a linear map $\rho : \text{Sym}^r V \rightarrow K$ such that

$$G = (\gamma, \rho) : \text{Sym}^r V \xrightarrow{\sim} S \oplus K$$

is an isomorphism. For the generic endomorphism M ,

$$\det(\text{Sym}^r M) = (\det M)^{\kappa(e, r)}, \quad \kappa(e, r) = \frac{r}{e} \binom{e+r-1}{r} = \binom{e+r-1}{r-1}.$$

The eigenvalue identity proves the formula on diagonal matrices and hence as a polynomial identity on $\text{End}(V)$.

Expand

$$\det(G \text{Sym}^r M) = \det(G) d^{\kappa(e,r)}$$

by Laplace expansion along the first s rows. Every summand is an s -minor of $\gamma \text{Sym}^r M$ times a complementary minor of $\rho \text{Sym}^r M$. Therefore the determinant belongs to $J_{\gamma,r}$. Since $\det G \neq 0$, $d^{\kappa(e,r)} \in J_{\gamma,r}$.

If γ is an isomorphism, the ideal of maximal minors is principal and generated by $\det(\gamma) \det(\text{Sym}^r M)$, so the exponent is sharp. $\square$

10.2. The depth bound and graded multiplication.

Proof of Theorem 6.2. Apply Theorem 6.1 with $e = 4$, $r = 2$, $\gamma = \gamma_R$, and $M = I_H \oplus T$. Since $\kappa(4, 2) = 5$, $d^5 \in I_6(\gamma_R \text{Sym}^2 M) = J$. Now $\mathcal{I}_{\widehat{D}_R} = dJ$. Therefore $d^6 \in dJ$, so the nilradical generated by the class of d satisfies $N^6 = 0$. Formula (9.4) then gives $W_j = \emptyset$ for $j \geq 6$.

For the multiplication statement work locally in $A = P/(dJ)$. The class of rd^i in N^i/N^{i+1} depends on r modulo

$$(d) + (J : d^{i-1}),$$

and multiplication sends

$$(rd^i)(sd^j) = rs d^{i+j}.$$

Because

$$(J : d^{i-1}) + (J : d^{j-1}) \subset (J : d^{i+j-1}),$$

this factors through the canonical restriction to W_{i+j} . The image of $1 \otimes 1$ is the generator d^{i+j} , hence the map is surjective. The transition functions of d supply the factor $\mathcal{L}^{\otimes(i+j)}$, proving (6.4). $\square$

10.3. The first layer at projection corank two. Assume throughout this subsection that A_H is injective. After eliminating its image, the residual ideal is the ideal of 3-minors of

$$\Psi_H(T) = \left[\beta_H(1_H \otimes T), \chi_H \text{Sym}^2 T \right], \quad \dim E_H = 3. \quad (10.1)$$

Lemma 10.1 (The contact symbol without mixed regularity). *Modulo (δ) , the residual ideal of (10.1) is generated in degree four and equals*

$$h_H(u) \text{Sym}^4(\mathbb{C}_v^2)$$

under $T = uv^t$. In particular

$$J + (\delta) = (\delta, g_0, \dots, g_4)$$

with the notation of Theorem 6.4.

Proof. On the rank-one cone write $T = uv^t$. The four mixed columns are scalar multiples, with coefficients v_1, v_2 , of the two vectors

$$\beta_H(h_1 \otimes u), \quad \beta_H(h_2 \otimes u),$$

and the three quadratic columns are scalar multiples, with coefficients $v_1^2, v_1 v_2, v_2^2$, of the single vector $\chi_H(u^2)$. A 3-minor containing three mixed columns vanishes because the mixed span has dimension at most two. A minor containing at least two quadratic columns vanishes because those columns are proportional. Thus only minors with two mixed columns and one quadratic column survive. Their common u -factor is

$$\det[\beta_H(h_1 \otimes u), \beta_H(h_2 \otimes u), \chi_H(u^2)],$$

a binary quartic. The five possible v -monomials span $\text{Sym}^4(\mathbb{C}_v^2)$. By the same determinant comparison used in Lemma 8.4, this binary quartic is the restriction h_H of the Jacobian quartic to the pencil of hyperplanes containing H . This proves the assertion without a rank assumption on β_H . $\square$

The orbit classification in Definition 6.3 is elementary projective geometry. We record normal forms because they are also used in the colon calculation. Order the basis of $H \otimes W$ as

$$h_1w_1, h_1w_2, h_2w_1, h_2w_2.$$

Up to $GL(H) \times GL(W) \times GL(E_H)$, the mixed block can be taken to be one of the following matrices:

type	β
3, off Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
3, on Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
2, secant	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
2, tangent	$\begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
2, H -ruling	$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
2, W -ruling	$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$
1, dual rank 2	$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$
1, dual rank 1	$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
0	0.

(10.2)

For rank two, a projective kernel line meets the smooth quadric in two reduced points, one double point, or is contained in one of the two ruling families. For rank one, the dual generator has tensor rank two or one. These alternatives are invariant and exhaustive, proving the orbit statement.

Lemma 10.2 (Exact transverse models for the nine mixed-kernel orbits). *Let*

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For each normal form in (10.2) there is a nonempty Zariski-open set of quadratic blocks χ on which the Hilbert functions of

$$P/((\delta) + (J : \delta^{j-1}))$$

are those displayed in Theorem 6.4.

Proof. The two rank-three rows are Theorem 9.3 and Theorem 9.5. For the remaining rows, the assertion is a homogeneous determinant calculation. We give integer witnesses and the resulting colons; this simultaneously proves nonemptiness of the relevant rank opens.

For the secant normal form take

$$\chi = \begin{pmatrix} 1 & 2 & -2 \\ -2 & -1 & 1 \\ -2 & 1 & -2 \end{pmatrix}.$$

A direct expansion of the 3-minors followed by cancellation by δ gives

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - c^2, ab - cd, b^2 - d^2, \\ ac - 2c^2, bc - 2cd, ad - 2cd, bd - 2d^2),$$

and $(J : \delta^2) = P$. The first ideal contains δ , and the quotient has Hilbert function $(1, 4, 3)$, hence length eight.

For the tangent normal form take

$$\chi = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ -1 & -2 & 2 \end{pmatrix}.$$

Then

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - 10c^2, ab - 10cd, b^2 - 10d^2, \\ ac + 2c^2, bc + 2cd, ad + 2cd, bd + 2d^2),$$

again with $(J : \delta^2) = P$. The same Hilbert function $(1, 4, 3)$ follows.

For the H -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{pmatrix}.$$

One obtains

$$(J : \delta) = (c^4, c^3d, c^2d^2, cd^3, d^4, 5ac^2 - c^3, 5bc^2 - c^2d, \\ 5acd - c^2d, 5bcd - cd^2, 5ad^2 - cd^2, 5bd^2 - d^3, \\ a^2 - ac, 2ab - ad - bc, b^2 - bd),$$

while

$$(J : \delta^2) = (a, b, c, d).$$

Here $\delta \notin (J : \delta)$, so $\delta^2 \notin J$; on the other hand $\delta \in (J : \delta^2)$, so $\delta^3 \in J$. After adjoining δ , the second-layer quotient has Hilbert function

$$(1, 4, 6, 4)$$

and length fifteen, while the third layer has length one.

For the W -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -2 \end{pmatrix}.$$

Then

$$(J : \delta) = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3), \quad (J : \delta^2) = P.$$

This ideal already contains δ . The Hilbert function of the second layer is

$$1, 4, 6, 4, 5, 6, 7, 8, \dots,$$

hence its Hilbert polynomial is $n + 1$ from degree four onward.

For the dual-rank-two rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -2 \\ -2 & 0 & -1 \end{pmatrix}.$$

After adjoining δ , a homogeneous basis for the second colon may be chosen as

$$(\delta, a^2c, ac^2, a^2d, abd, b^2d, acd, ad^2, bd^2).$$

Its Hilbert function is

$$1, 4, 9, 8, 10, 12, 14, 16, \dots,$$

so the Hilbert polynomial is $2n + 2$ from degree three onward. Moreover

$$(J : \delta^2) = (a, b, c, d),$$

and the third layer is the reduced vertex.

For the dual-rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 2 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

A homogeneous basis after adjoining δ is

$$(\delta, a^3 + 4a^2c + 3ac^2, a^2b + 4a^2d + 3acd, \\ ab^2 + 4abd + 3ad^2, b^3 + 4b^2d + 3bd^2),$$

with Hilbert function

$$1, 4, 9, 12, 15, 18, 21, \dots$$

The next colon is, after an invertible linear change,

$$(c^2, cd, d^2, a, b),$$

so the third layer has Hilbert function $(1, 2)$ and length three.

Finally, when $\beta = 0$, surjectivity of $[\beta, \chi]$ forces χ to be invertible. Hence

$$J = I_3(\chi \operatorname{Sym}^2 T) = (\det \chi) \det(\operatorname{Sym}^2 T) = (\delta^3),$$

because $\det(\operatorname{Sym}^2 T) = \delta^3$ in dimension two. This proves the last row exactly.

All displayed Hilbert functions are computed from the displayed homogeneous bases by counting standard monomials. The role of these integral points is only to provide convenient base points. Genericity is proved separately in Theorem 11.1: the stabilizer orbit of each normal form is supplemented by an explicit transverse affine slice, and the residual colons are computed over the rational function field of that slice. The resulting initial ideals give exactly the Hilbert functions displayed above. Hence the open conditions in the statement are genuine dense-open conditions on the whole orbit family, not constant-rank loci inferred from one fibre. $\square$

Proof of Theorem 6.4. Lemma 10.1 gives the first layer and its contact interpretation. The orbit list follows from (10.2). Lemma 10.2 gives the second and third layers on each coefficient-transverse open. The nilpotency index is

$$1 + \min\{k : \delta^k \in J\}$$

by Theorem 9.1; the displayed colons give the values in the table.

If $b \leq 1$, two mixed columns never span two independent target directions, so the determinant in the proof of Lemma 10.1 vanishes identically and $h_H = 0$. If the kernel is the H -ruling $h_0 \otimes W$, one of the two mixed vectors vanishes for every u , so again $h_H = 0$. For the W -ruling, the mixed span drops only at $u = w_0$, giving the forced contact root but not identical vanishing on the transverse open. This proves the remaining assertions. $\square$

10.4. Rank-drop depth and the finite coefficient atlas.

Proof of Proposition 6.8. Eliminate the rank- a block A_H and put $q = 6 - a$. The remaining target has dimension q . A maximal residual minor uses at most b independent mixed columns, each of transverse degree one. Every remaining column must come from the quadratic block and has transverse degree two. Therefore every nonzero generator of J has degree at least

$$b + 2(q - b) = 2q - b.$$

If $2k < 2q - b$, the homogeneous polynomial δ^k cannot lie in J , proving

$$k \geq q - \lfloor b/2 \rfloor.$$

Theorem 6.2 gives $k \leq 5$.

The quadratic block has only three source dimensions when $c = 2$. Surjectivity of the original quotient after eliminating A_H therefore implies $b + 3 \geq q$. Combining this with $b \leq 4$ gives exactly the table in the statement. The equivalence between $d^k \in J$ and the augmented-rank condition $\varepsilon_k(J) = 0$ is ordinary linear algebra in the degree $2k$ homogeneous component. In the type $(a, b) = (0, 3)$ the lower bound already gives $k \geq 5$, while $d^5 \in J$, so $k = 5$ exactly. $\square$

The same coefficient construction works for $c = 3, 4$. Because $d^5 \in J$, it terminates at $k = 5$. Colon ideals are kernels of the maps

$$(P_c)_m \xrightarrow{\cdot d^{j-1}} (P_c)_{m+c(j-1)} / J_{m+c(j-1)},$$

hence are obtained by finite syzygy elimination. Noetherianity gives finite homogeneous generating sets. Rank stratification of the resulting coefficient matrices, followed by the usual flattening stratification of the finitely presented graded quotients, gives the finite atlas in Definition 6.5. This is the finite algebraic replacement for an unspecified infinite colon tower.

10.5. Projection corank three.

Lemma 10.3 (Functorial symbol containment under mixed-rank drop). *Fix the rank of A_H at projection corank three. In every homogeneous degree, the right $GL(W)$ -module generated by the residual maximal minors for a specialization of the mixed block is a quotient of the corresponding universal maximal-mixed-rank symbol module. After multiplication by coordinate functions the same is true degree by degree.*

Proof. After eliminating in A_H , a residual minor with u mixed columns and $q - u$ quadratic columns is the image of the natural alternating map

$$\bigwedge^u (H \otimes W) \otimes \bigwedge^{q-u} \text{Sym}^2 W \longrightarrow \bigwedge^q E_H.$$

The construction is polynomial and functorial in the mixed map $H \otimes W \rightarrow E_H$. Replacing that map by one of smaller rank factors the displayed map through the corresponding quotient of $H \otimes W$; it cannot create a new right $GL(W)$ -constituent. Tensoring with homogeneous coordinate functions and taking images preserves this quotient relation. $\square$

Lemma 10.4 (No cubic determinant power at corank three). *At projection corank three, $d^3 \notin J$.*

Proof. Here $\dim W = 3$ and $\deg d = 3$. First suppose A_H has rank one. After eliminating its image the residual target has dimension five. The maximal-rank mixed case has rank $\overline{B}_H = 3$. By Lemma 10.3, every mixed-rank drop is a quotient of the symbol module treated below, degree by degree. Thus a determinant representation absent from the universal symbol cannot be created by specialization.

The degree-seven, degree-eight and degree-nine symbols of the five-minors use respectively three, two and one mixed columns. As right $GL(W)$ -representations they are contained in

$$\bigwedge^3 W \otimes \bigwedge^2 \text{Sym}^2 W, \quad \bigwedge^2 W \otimes \bigwedge^3 \text{Sym}^2 W, \quad W \otimes \bigwedge^4 \text{Sym}^2 W.$$

The elementary Schur decompositions are

$$\begin{aligned} \bigwedge^2 \text{Sym}^2 W &= \mathbb{S}_{(3,1,0)} W, \\ \bigwedge^3 \text{Sym}^2 W &= \mathbb{S}_{(4,1,1)} W \oplus \mathbb{S}_{(3,3,0)} W, \end{aligned}$$

and

$$\bigwedge^4 \text{Sym}^2 W = \mathbb{S}_{(4,3,1)} W.$$

Consequently the degree-eight symbol is a sum of

$$\mathbb{S}_{(5,2,1)}, \mathbb{S}_{(4,4,0)}, \mathbb{S}_{(4,3,1)}, \mathbb{S}_{(4,2,2)},$$

and the degree-nine symbol is a sum of

$$\mathbb{S}_{(5,3,1)}, \mathbb{S}_{(4,4,1)}, \mathbb{S}_{(4,3,2)}.$$

The degree-seven symbol is $\mathbb{S}_{(4,2,1)}$. To obtain a degree-nine element of the ideal one may multiply the degree-seven symbol by degree-two coordinate functions, the degree-eight symbol by degree-one coordinate functions, or take the degree-nine symbol itself. None can contain

$$(\det W)^3 = \mathbb{S}_{(3,3,3)} W :$$

every displayed highest weight has first part at least four, and the Littlewood–Richardson rule only adds boxes. Since d^3 spans this determinant representation, it is not in J .

If $A_H = 0$, the residual target has dimension six. A degree-nine minor can occur only when the mixed block has rank three and uses all three mixed columns and three quadratic columns. Its right type is contained in

$$\det W \otimes \bigwedge^3 \text{Sym}^2 W = \mathbb{S}_{(5,2,2)} W \oplus \mathbb{S}_{(4,4,1)} W,$$

again with no $(3,3,3)$ summand. If the mixed rank drops, the minimal degree is larger than nine. This proves the assertion in all cases. $\square$

Proof of Theorem 6.9. Lemma 10.4 gives $d^3 \notin J$, while Theorem 6.2 gives $d^5 \in J$. Therefore the first determinant power in J is d^4 or d^5 . Definition 6.5 says precisely that $d^4 \in J$ when $\varepsilon_4(J) = 0$. The nilpotency index is one plus this first exponent. $\square$

10.6. Projection corank four. We use one standard plethysm calculation (see, for example, [18] for Schur functors and character methods). The identity may also be checked directly by the Weyl character formula:

$$\bigwedge^6 \text{Sym}^2 V = \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (10.3)$$

Both summands occur with multiplicity one.

Lemma 10.5 (Equivariance of the corank-four residual ideal). *Let $A = \text{Sym}(V^* \otimes V) = \mathbb{C}[\text{End}(V)]$ and $J = I_6(\gamma_R \text{Sym}^2 T)$ on the projection-corank-four chart. For the right action $T \mapsto Tg$ of $GL(V)$, the ideal J is stable. Hence every homogeneous piece $J_m \subset A_m$ is a $GL(V)$ -subrepresentation.*

Proof. For $g \in GL(V)$,

$$\gamma_R \text{Sym}^2(Tg) = \gamma_R(\text{Sym}^2 T)(\text{Sym}^2 g).$$

The last factor is invertible. Maximal minors are therefore changed only by an invertible linear combination of the columns, so their ideal is unchanged. Homogeneity gives the assertion degree by degree. $\square$

Lemma 10.6 (Effective Cauchy–Pieri multiplication in the matrix ring). *Let $A = \text{Sym}(V^* \otimes V) = \mathbb{C}[\text{End}(V)]$. In the Cauchy decomposition of A , put*

$$\lambda = (4, 4, 4, 0), \quad \mu = (4, 0, 0, 0), \quad \nu = (4, 4, 4, 4).$$

The actual polynomial multiplication

$$A_{12}[\lambda] \otimes A_4[\mu] \longrightarrow A_{16}$$

has nonzero projection to $A_{16}[\nu]$. This projection has multiplicity one on both the left and right Schur factors and, after the identifications

$$\mathbb{S}_\lambda V \simeq (\det V)^4 \otimes (\text{Sym}^4 V)^*, \quad \mathbb{S}_\mu V = \text{Sym}^4 V,$$

is a nonzero scalar multiple of the evaluation pairing.

Proof. The skew diagram ν/λ consists of four boxes, one in each column, hence is a horizontal four-strip and $c'_{\lambda,\mu} = 1$. We also record a coordinate-ring nonvanishing, because abstract Littlewood–Richardson occurrence is not sufficient here.

Let $X = (x_{ij})$ be the generic 4×4 matrix and use the standard bitableau realization of the Cauchy decomposition. Write $[123 \mid 123]$ for the north-west 3×3 minor and $[4 \mid 4] = x_{44}$. The standard monomials

$$[123 \mid 123]^4 \in A_{12}[\lambda], \quad [4 \mid 4]^4 \in A_4[\mu]$$

represent the extremal weight vectors adapted to the unique Pieri tableau that adds one box to the bottom of each of the four columns of λ . Standard bitableau straightening gives

$$[123 \mid 123]^4 [4 \mid 4]^4 = [1234 \mid 1234]^4 + \sum_{\tau \neq \nu} c_\tau B_\tau,$$

where the coefficient of the unique ν -standard bitableau is 1. Since $[1234 \mid 1234]^4 = (\det X)^4$, the ν -projection of actual polynomial multiplication is nonzero. This is the standard straightening coefficient attached to the unique horizontal four-strip; see [8].

The relevant Littlewood–Richardson multiplicity is one, so Schur’s lemma identifies the induced maps on the left and right Schur factors with the unique Pieri maps up to nonzero scalar. Under

$$\mathbb{S}_\lambda V \simeq (\det V)^4 \otimes (\mathrm{Sym}^4 V)^*, \quad \mathbb{S}_\mu V = \mathrm{Sym}^4 V,$$

the right-factor map is therefore the perfect evaluation pairing up to a nonzero scalar. $\square$

Lemma 10.7 (The Jacobian projection). *Under*

$$\bigwedge^6 \mathrm{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \mathrm{Sym}^2 V \right)^*,$$

the summand $\mathbb{S}_{(4,4,4,0)} V$ is dual, up to the displayed determinant twist, to

$$\det V \otimes \mathrm{Sym}^4 V \subset \bigwedge^4 \mathrm{Sym}^2 V.$$

The projection

$$\bigwedge^4 \mathrm{Sym}^2 V \longrightarrow \det V \otimes \mathrm{Sym}^4 V$$

sends $q_1 \wedge \cdots \wedge q_4$, up to a nonzero scalar, to

$$\det \left(\frac{\partial q_i}{\partial x_j} \right).$$

Proof. Duality of Schur functors gives

$$(\det V)^5 \otimes \mathbb{S}_{(4,4,4,0)} V^* \simeq \det V \otimes \mathrm{Sym}^4 V.$$

The Jacobian determinant is alternating in the four quadrics and is $GL(V)$ -equivariant with exactly this target weight. It is nonzero, for instance on $x_1^2, x_2^2, x_3^2, x_4^2$. Multiplicity one in (10.3) therefore identifies it with the stated projection. $\square$

Proof of Theorem 6.10. At $H = 0$ the ideal J is generated in degree twelve by the 6-minors of $\gamma_R \mathrm{Sym}^2 T$. Under right multiplication of T , their span is a quotient of the representation $\bigwedge^6 \mathrm{Sym}^2 V$. The polynomial d^3 spans $(\det V)^3 = \mathbb{S}_{(3,3,3,3)} V$, which does not occur in (10.3). Hence $d^3 \notin J$.

Let

$$\lambda_R = \bigwedge^6 \gamma_R.$$

By Lemma 10.7, the component of λ_R on the $\mathbb{S}_{(4,4,4,0)}$ -summand is nonzero exactly when the Jacobian quartic F_R is nonzero. For $R \in G_4^\circ$, the quartic $Y_R = V(F_R)$ is smooth, so this component is nonzero.

Let $A = \mathbb{C}[\text{End}(V)]$. The nonzero Jacobian component just identified gives a nonzero vector in the $\lambda = (4, 4, 4, 0)$ Cauchy component of J_{12} . By Lemma 10.6, multiplication by the $\mu = (4)$ component of A_4 has a nonzero $\nu = (4, 4, 4, 4)$ -projection. Because the induced pairing

$$\mathbb{S}_{(4,4,4,0)} V \otimes \text{Sym}^4 V \longrightarrow (\det V)^4$$

is nondegenerate, one may choose the degree-four matrix coefficient to pair nontrivially with the specific nonzero Jacobian vector.

By Lemma 10.5, J_{16} is a right $GL(V)$ -subrepresentation of A_{16} . Over $\mathbb{C}$ it is completely reducible, so the equivariant projection onto any right isotypic summand preserves J_{16} . Apply the projection onto right type $\nu = (4, 4, 4, 4)$ to the product just constructed. Its image is nonzero by Lemma 10.6 and still belongs to J_{16} .

In the Cauchy decomposition of the full matrix coordinate ring, right type ν is paired with the same left type. Both $\mathbb{S}_{(4,4,4,4)} V$ and $\mathbb{S}_{(4,4,4,4)} V^*$ are one-dimensional determinant characters. Hence this entire Cauchy summand is the one-dimensional line

$$\mathbb{C}(\det T)^4 = \mathbb{C}d^4.$$

A nonzero element of this line lying in J_{16} therefore implies

$$d^4 \in J.$$

Thus ideal membership follows from three explicit ingredients: equivariance of the maximal-minor ideal, semisimple projection in characteristic zero, and the nonzero coordinate-ring Pieri coefficient. No projection is taken outside the ideal. Together with $d^3 \notin J$, this gives nilpotency index five. $\square$

10.7. Specialization and associated supports. The coefficient matrices above have polynomial entries, but this fact alone does not justify fibrewise statements about embedded associated points or base change of colons. The required order of argument is carried out in Section 11. There we first flatten the finite presentations controlling each kernel ($J : d^{j-1}$) and its cokernel, then apply relative assassins, generic freeness at the associated generic points, and Noetherian induction. Proposition 6.7 follows from that argument.

The same section computes the generic corank-two primary signatures over rational function fields and gives explicit one-parameter orbit degenerations. Thus the specialization statements used by the paper are consequences of exact family ideals and relative associated-point theory, not of generic flatness alone.

11. RELATIVE PRIMARY SIGNATURES AND SPECIALIZATION LAWS

This section supplies the two family arguments that are not consequences of a single exact witness: genericity on coefficient orbits and fibrewise control of associated points. It also extracts the geometric primary signatures of the corank-two layers and writes explicit orbit degenerations.

11.1. Stabilizer slices and generic Groebner bases. Keep

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For a mixed normal form β , let $G_\beta \subset GL(H) \times GL(W) \times GL(E_H)$ be its stabilizer. If (X, Y, Z) lies in its Lie algebra, then

$$Z\beta - \beta(X \otimes 1 + 1 \otimes Y) = 0, \tag{11.1}$$

and its infinitesimal action on the quadratic block is

$$\dot{\chi} = Z\chi - \chi \operatorname{Sym}^2 Y. \quad (11.2)$$

These are linear equations over $\mathbb{Q}$ at the integral base points of Lemma 10.2.

Theorem 11.1 (Generic slice certification for the mixed-kernel atlas). *For each of the seven mixed-kernel rows not already covered by the mixed-regular and decomposable-kernel theorems, the Hilbert functions in Theorem 6.4 are the generic Hilbert functions on the entire orbit family.*

More precisely, at the integral base points used in Lemma 10.2, the ranks of the infinitesimal G_β -orbits in the nine-dimensional χ -space and one choice of transverse affine slice are

type	$\operatorname{rank} T_\chi(G_\beta\chi)$	slice entries
secant	6	$\chi_{11}, \chi_{21}, \chi_{31}$
tangent	7	χ_{11}, χ_{21}
H -ruling	7	χ_{11}, χ_{13}
W -ruling	9	$\emptyset$
dual rank 2	9	$\emptyset$
dual rank 1	9	$\emptyset$
zero mixed map	9	$\emptyset$.

On the first three slices, with coordinates s_i , Groebner bases of all required colons over $\mathbb{Q}(s_i)$ give respectively

type	$\min\{k : \delta^k \in J\}$	H_{W_2}	H_{W_3}
secant	2	$(1, 4, 3)$	0
tangent	2	$(1, 4, 3)$	0
H -ruling	3	$(1, 4, 6, 4)$	(1) .

The only nonconstant denominators in reduced bases may be chosen to divide

$$s_2 - 2 \quad (\text{secant}), \quad s_0 + 2s_1 - 5 \quad (H\text{-ruling}),$$

and neither vanishes at the base point $s = 0$; the tangent slice requires no nonconstant denominator. In the remaining four rows the stabilizer orbit is already open in χ -space, so the exact base-point calculation is the generic calculation.

Proof. Solving (11.1) and applying (11.2) gives the ranks in the table. Adjoining the displayed coordinate directions raises the tangent rank to nine. Thus the morphism from the stabilizer times the affine slice to χ -space has surjective differential at the base point. Over characteristic zero it is therefore dominant and smooth on a neighbourhood of that point.

Compute the residual minors and successive colons in $\mathbb{Q}(s_i)[a, b, c, d]$. For the secant and tangent slices the second-layer generic initial ideal is

$$(c, d)^3 + (a, b)^2 + (a, b)(c, d),$$

which has Hilbert function $(1, 4, 3)$, and the next colon is the unit ideal. For the H -ruling slice a generic initial ideal of the second layer is generated by

$$c^4, c^3d, c^2d^2, cd^3, d^4, ac^2, acd, ad^2, bd^2, a^2, ab, b^2, bc,$$

with Hilbert function $(1, 4, 6, 4)$; the third-layer initial ideal is (a, b, c, d) . These initial ideals are unchanged after specialization away from the displayed denominator divisors.

For the four open-orbit rows the exact Groebner bases give the functions printed in Theorem 6.4. The calculation is reproduced, over exact rational function fields rather than floating-point samples, by `checks/generic_boundary_atlas.py`. Since the slice map is dominant, these are the generic initial ideals on the orbit families. $\square$

11.2. Geometric primary signatures for the nine mixed-kernel types. Let $Q = V(\delta)$ be the affine Segre cone and let $\mathfrak{m} = (a, b, c, d)$ be its vertex. Write R_r for a reduced divisor consisting of r distinct lines in one ruling of Q . Write V_ℓ for a homogeneous $\mathfrak{m}$ -primary scheme of length ℓ . Finally, $R_r + E_\ell$ means a scheme whose saturation is R_r and whose saturation quotient is an embedded vertex module of length ℓ . This notation records associated supports and generic lengths; it does not choose a noncanonical embedded primary component.

Theorem 11.2 (Generic primary signatures for the mixed-kernel orbits). *On the generic coefficient stratum of each A_H -injective mixed-kernel orbit, the positive layers have the following primary signatures:*

<i>mixed-kernel type</i>	W_2	W_3
3, point off Q	V_1	$\emptyset$
3, point on Q	V_3	$\emptyset$
2, secant	V_8	$\emptyset$
2, tangent	V_8	$\emptyset$
2, H -ruling	V_{15}	V_1
2, W -ruling	$R_1 + E_5$	$\emptyset$
1, dual rank 2	$R_2 + E_3$	V_1
1, dual rank 1	R_3	V_3
0	Q	Q .

In particular, every associated support and every generic length in these layers is determined. The symbols R_r have r minimal ruling-line primes, each of generic length one. The only additional associated prime in $R_1 + E_5$ and $R_2 + E_3$ is the embedded vertex $\mathfrak{m}$.

Proof. The finite rows are homogeneous zero-dimensional schemes supported at the vertex, hence are $\mathfrak{m}$ -primary; their lengths are the Hilbert-function sums. For the W -ruling witness the second layer is

$$I = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3).$$

Its saturation is (c, d) , and $(c, d)/I$ has length five. Thus its associated primes are (c, d) and $\mathfrak{m}$, with an embedded vertex contribution of length five.

For the dual-rank-two witness, saturation gives

$$(a, b) \cap (c, d),$$

the union of two ruling lines, and the saturation quotient has length three. For the dual-rank-one witness, modulo δ , the second-layer ideal is exactly

$$\bigcap_{\lambda \in \{0,1,3\}} (a + \lambda c, b + \lambda d),$$

so it is a reduced union of three ruling lines and has no embedded point. The third layer is the length-three vertex thickening already computed in Lemma 10.2. The zero mixed-map identity $J = (\delta^3)$ gives $W_2 = W_3 = Q$.

Theorem 11.1 transports these exact signatures to the generic coefficient strata. Associated supports are preserved by the stabilizer action, and the relative-primary argument below prevents an isolated base-point calculation from being used beyond its certified open stratum. $\square$

11.3. Pure-block rank drop at projection corank two. When A_H has rank $a < 3$, put $q = 6 - a$ and $b = \text{rank } \overline{B}_H$. For $b = 3, 2, 1$ the phrase *generic mixed kernel* means respectively an off-Segre kernel point, a secant kernel line, and a kernel plane whose dual tensor has rank two.

Theorem 11.3 (Generic primary signatures for every A_H -rank-drop type). *At projection corank two, every generic rank type allowed by Proposition 6.8 has the following complete list of nonzero positive layers:*

a	b	$nilpotency\ index$	$(W_1, W_2, W_3, W_4, W_5)$
2	4	3	(Q, V_{22})
2	3	4	(Q, V_{27}, V_1)
2	2	4	(Q, R_4, V_8)
2	1	5	(Q, Q, Q, V_1)
1	4	5	(Q, Q, V_{11}, V_1)
1	3	5	$(Q, Q, R_2 + E_3, V_1)$
1	2	5	(Q, Q, Q, R_2)
0	4	6	(Q, Q, Q, R_2, V_1)
0	3	6	(Q, Q, Q, Q, V_1) .

Here every Q is the reduced Segre quadric with generic length one; every R_r is a reduced union of r ruling lines with generic length one on each component; and every V_ℓ is $\mathfrak{m}$ -primary of the indicated length. In the unique row $(a, b) = (1, 3)$ carrying E_3 , the associated primes of that layer are the two ruling-line primes together with the embedded vertex.

Proof. Choose the generic normal forms of the mixed block described above and let χ vary. The stabilizer-slice codimensions in χ -space for

$$(a, b) = (2, 4), (2, 3), (2, 2), (2, 1), (1, 4), (1, 3), (1, 2), (0, 4), (0, 3)$$

are respectively

$$5, 4, 1, 0, 3, 1, 0, 0, 0.$$

On explicit complementary coordinate slices the successive colons are computed over the rational function fields of those slices. Their Hilbert functions are

a	b	non-quadric Hilbert functions
2	4	$(1, 4, 9, 8)$
2	3	$(1, 4, 9, 8, 5); (1)$
2	2	$4n + 4 \ (n \geq 4); (1, 4, 3)$
2	1	(1)
1	4	$(1, 4, 6); (1)$
1	3	$2n + 2 \ (n \geq 3); (1)$
1	2	$2n + 2 \ (n \geq 1)$
0	4	$2n + 2 \ (n \geq 1); (1)$
0	3	$(1).$

Every omitted earlier layer has Hilbert function $(n + 1)^2$, hence equals Q .

To identify the positive-dimensional supports, pull the layer ideals back to the rank-one parametrization $T = uv^t$, but now perform the factorization over the same rational-function fields as the Groebner calculation.

For $(a, b) = (2, 2)$, with slice parameter s , the second layer has common rank-one factor, up to a unit of $\mathbb{Q}(s)$,

$$u_0 u_1 \left((s + 9)u_0^2 + (3s + 6)u_0 u_1 - 7u_1^2 \right).$$

The quadratic discriminant is

$$9s^2 + 64s + 288.$$

On the open set $(s + 9)(9s^2 + 64s + 288) \neq 0$, its two roots are distinct and different from the two roots $u_0 u_1 = 0$. Thus the generic support is a reduced divisor of four distinct ruling lines. Its Hilbert polynomial agrees with the computed layer, so there is no generic embedded correction.

For $(a, b) = (1, 3)$, with slice parameter t , the third layer has common factor, again up to a unit of $\mathbb{Q}(t)$,

$$q(u) = 10u_0^2 + u_0u_1 - 4u_1^2.$$

Its discriminant is 161, so it gives two distinct ruling lines over the algebraic closure of $\mathbb{Q}(t)$. The exact generic-field calculation also computes the saturation. Writing $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the saturated ruling ideal is

$$I_{\text{sat}} = (\delta, 10a^2 + ac - 4c^2, 10ab + ad - 4cd, \\ 10b^2 + bd - 4d^2).$$

The Groebner bases over $\mathbb{Q}(t)$ give

$$H_{M_3} = (1, 4, 9, 8, 10, 12, 14, \dots), \quad H_{P/I_{\text{sat}}} = (1, 4, 6, 8, 10, 12, 14, \dots).$$

The quotient I_{sat}/I_{M_3} is therefore concentrated in degree two and has length three. This is the claimed embedded vertex module, so the generic signature is $R_2 + E_3$.

For the two open-orbit rows $(1, 2)$ and $(0, 4)$, the previously displayed base-point factors remain generic by orbit transport:

$$u_0u_1, \quad 6u_0^2 + u_0u_1 + 4u_1^2,$$

and the second quadratic has nonzero discriminant. Thus every positive-dimensional rank-drop support in the table is certified on its generic coefficient field, rather than inferred from an isolated integral point. The exact stabilizer ranks, slice coordinates, Groebner leading monomials, denominator checks, and Hilbert counts are reproduced by `checks/generic_boundary_atlas.py`. Because each slice map is dominant at its base point, the calculation is at the generic point of the rank type rather than at a special witness. $\square$

11.4. Relative assassins and base change for the colon tower. We prove the general bounded principal-colon theorem before applying it to the determinant d . The order of the argument is essential.

Lemma 11.4 (Finite geometric-assassin stratification). *Let $X \rightarrow S$ be of finite presentation with S noetherian, and let $\mathcal{F}$ be a finitely presented coherent module which is flat over S . After a finite locally closed stratification of S , there are finitely many closed support packets $Z_\nu \subset X$ such that for every geometric point $\bar{s} \rightarrow S$,*

$$\text{Ass}(\mathcal{F}_{\bar{s}}) = \bigcup_{\nu} \{\text{generic points of irreducible components of } (Z_\nu)_{\bar{s}}\}.$$

The packets may split geometrically. After further finite refinement, their dimensions, incidence relations, minimal-versus-embedded status, and the numbers

$$\text{length } H_{p\mathcal{O}_{X_{\bar{s}},p}}^0(\mathcal{F}_{\bar{s},p})$$

at their associated generic points are constant.

Proof. Work componentwise on the base and use Noetherian induction. Over a geometric generic point the associated set is finite. After a finite extension of the generic field, collect Galois-conjugate associated points into finitely many packets, take their closures in X , and shrink so that these closures are flat and their geometric fibre dimensions are constant. The base-change description of the relative assassin in [21, Tag 05AS] permits this geometric formulation. For a flat module, the generic points of fibres of the closure of a relative associated point remain relative associated points after shrinking; we use [22, Tag 0GSJ].

What remains is to exclude untracked fibrewise associated points. The needed locus is constructible, not merely specialization closed. We use the relative-assassin constructibility theorem [24, Tag 05KR]: for a quasi-compact open $U \subset X$, the locus of $s \in S$ for which

$\text{Ass}(\mathcal{F}_s) \subset U_s$ is locally constructible. Applying this to a finite affine cover adapted to the closures just constructed, and intersecting the resulting constructible conditions, gives a dense constructible locus on which the generic packets exhaust the geometric fibrewise assassin. A dense constructible subset of a noetherian irreducible scheme contains a nonempty open. Remove that open and repeat on the noetherian complement. This terminates and gives a finite stratification. The same constructibility argument applied to pairwise closures fixes incidence and therefore minimal-versus-embedded status.

For the multiplicity assertion, fix a tracked generic point p . The module

$$T_p = H_{p\mathcal{O}_{X,p}}^0(\mathcal{F}_p)$$

has finite length. Choose n with $p^n T_p = 0$, spread both the packet ideal and the finite torsion module after shrinking, and filter it by the powers of that ideal. The successive quotients are finite modules on the packet. Generic freeness makes their generic ranks constant after shrinking, and the sum of those ranks is exactly the displayed length. Noetherian induction handles the complement. $\square$

Proof of Theorem 6.6. Fix $0 \leq q < N$ and put

$$B = A/J, \quad C_q = B/f^q B = A/(J, f^q), \quad I_q = f^q B.$$

There are two short exact sequences

$$0 \longrightarrow I_q \longrightarrow B \longrightarrow C_q \longrightarrow 0, \quad 0 \longrightarrow K_q \longrightarrow A \xrightarrow{a \mapsto af^q} I_q \longrightarrow 0. \quad (11.3)$$

There is also

$$0 \longrightarrow J \longrightarrow A \longrightarrow B \longrightarrow 0.$$

By generic flatness, simultaneously for the finite list $q = 0, \dots, N-1$, followed by Noetherian induction on the complement, refine S so that B and every C_q are flat. Then the first sequence in (11.3) shows that every I_q is flat. Since A is flat, the second sequence remains exact after *arbitrary* base change. Since B is flat, the sequence defining J does as well.

Let $T \rightarrow S_\alpha$ be any base change. Exactness identifies

$$B_T = A_T/J_T, \quad (I_q)_T = f_T^q B_T,$$

and the kernel of $A_T \rightarrow (I_q)_T$ is consequently

$$(K_q)_T = (J_T : f_T^q).$$

This is the desired colon base-change identity. Notice that flatness of the final cokernel C_q alone is not being used to preserve a kernel; the image module I_q has first been proved flat.

Now refine once more so that every $M_q = A/(f, K_q)$, and the sources, targets and cokernels of the finite list of multiplication maps under consideration, are flat. Their Hilbert polynomials and fibre ranks are then constant and commute with base change. Apply Lemma 11.4 to each M_q , and take a common finite refinement. This produces the support packets, incidences, minimal/embedded status and embedded multiplicities asserted in the theorem. $\square$

Proof of Proposition 6.7. Apply Theorem 6.6 to the finite-type coefficient chart with

$$A = P_S, \quad f = d, \quad N = 5.$$

The hypothesis $d^5 \in J$ is Theorem 6.2, itself a specialization of the universal symmetric-power determinant-completion theorem. Include the finitely many graded multiplication maps in the same refinement. The theorem gives base change for every residual colon, flatness and constant Hilbert polynomial for every $\mathcal{M}_j$, the exact geometric fibrewise associated-point packets with constant embedded multiplicities, and constant multiplication ranks. These are precisely the three clauses of the proposition.

The construction of the coefficient strata is noncanonical, but its fibrewise output is the intrinsic graded algebra, its geometric assassin, and the canonical torsion lengths at associated generic points. Two presentation atlases therefore admit a common refinement carrying the same primary signature. $\square$

11.5. Explicit orbit closures and graded specialization. Let

$$e_1 = h_1 w_1, \quad e_2 = h_1 w_2, \quad e_3 = h_2 w_1, \quad e_4 = h_2 w_2$$

be the standard basis of $H \otimes W$. The following families are written before applying a general quadratic block χ_t .

Theorem 11.5 (Corank-two orbit closure and specialization laws). *The closure relations needed to pass between all nine mixed-kernel types are realized by the following one-parameter families:*

- (1) *rank three, off Q to on Q :*

$$\beta_t = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -t & 0 & 0 & 1 \end{pmatrix}, \quad \ker \beta_t = \langle e_1 + te_4 \rangle;$$

- (2) *secant to tangent:*

$$\ker \beta_t = \langle e_1 + e_4, e_2 - te_3 \rangle,$$

for which $\delta|_{\mathbb{P}(\ker \beta_t)} = x^2 + ty^2$;

- (3) *secant to the two ruling families:*

$$\langle e_1, e_2 + te_4 \rangle \rightsquigarrow \langle e_1, e_2 \rangle, \quad \langle e_1, e_3 + te_4 \rangle \rightsquigarrow \langle e_1, e_3 \rangle;$$

- (4) *tangent to dual-rank-two:*

$$\beta_t = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & t \\ 0 & 0 & 0 & 0 \end{pmatrix};$$

- (5) *either ruling to dual-rank-one, by multiplying the second nonzero row in the corresponding rank-two normal form by t ;*

- (6) *either rank-one type to the zero map, by replacing β with $t\beta$.*

For a general polynomial choice of χ_t , the generic fibre and special fibre lie in the generic coefficient strata of the indicated orbit types. The corresponding primary signatures specialize as follows:

degeneration	change of positive layers
off $Q \rightarrow$ on Q	$V_1 \rightarrow V_3$
secant $\rightarrow$ tangent	$V_8 \rightarrow V_8$
secant $\rightarrow H$ -ruling	$V_8 \rightarrow (V_{15}, V_1)$
secant $\rightarrow W$ -ruling	$V_8 \rightarrow (R_1 + E_5)$
tangent $\rightarrow$ dual rank 2	$V_8 \rightarrow (R_2 + E_3, V_1)$
H -ruling $\rightarrow$ dual rank 1	$(V_{15}, V_1) \rightarrow (R_3, V_3)$
W -ruling $\rightarrow$ dual rank 1	$(R_1 + E_5) \rightarrow (R_3, V_3)$
dual rank 2 $\rightarrow$ 0	$(R_2 + E_3, V_1) \rightarrow (Q, Q)$
dual rank 1 $\rightarrow$ 0	$(R_3, V_3) \rightarrow (Q, Q)$.

Thus ruling components and embedded vertex components are born in actual one-parameter families of Fitting schemes.

Proof. The determinant of a tensor in each displayed kernel gives the stated Segre incidence. For example

$$\det \left(xI_2 + y \begin{pmatrix} 0 & 1 \\ -t & 0 \end{pmatrix} \right) = x^2 + ty^2,$$

which is secant for $t \neq 0$ and tangent for $t = 0$. Likewise δ restricts to txy on each secant-to-ruling family. The row-scaling families have the stated rank-one limits by inspection.

The residual maximal minors are polynomial in t and in the coefficients of χ_t . Each source and target generic stratum is a nonempty open in its coefficient space. Choosing χ_0 in the target open and a general first-order perturbation $\chi_0 + t\chi_1$ avoids the source bad divisors over $\mathbb{C}(t)$. Hence the two fibres have the primary signatures in Theorem 11.2. Because the ideals come from one polynomial family, the changes displayed in the table are genuine specializations, not comparisons of unrelated normal forms. $\square$

There is an analogous pure-block degeneration. In a fixed target basis let

$$A_t = (f_1, f_2, tf_3) : \text{Sym}^2 H \longrightarrow S_R$$

and choose the remaining mixed and quadratic coefficients generally. For $t \neq 0$, the induced mixed block is generically the rank-three off-Segre type; at $t = 0$, A_H has rank two and the induced mixed block has rank four. Thus the first nontrivial layer specializes from V_1 to V_{22} . This gives an explicit A_H -injective-to-rank-drop family, while the full table of Theorem 11.3 describes the generic pure-block boundary reached after further rank drops.

12. DEFORMATION TO THE NORMAL CONE: STANDARD MECHANISM AND THE COMPUTED SPECIAL FIBRE

The passage from an ideal to its extended Rees algebra is the standard deformation-to-the-normal-cone construction; see [17, §5.1] and [19, Tag 062Z]. We use it here only as the canonical mechanism that packages the manuscript-specific graded algebra computed in Sections 9 and 10.

Proposition 12.1 (Intrinsic deformation to the normal cone). *Let $\widehat{\mathcal{N}}_R$ be the intrinsic nilradical of $\widehat{D}_R$. On an affine open $\text{Spec } A \subset \widehat{D}_R$, with nilradical $N \subset A$, put*

$$\mathcal{R}_N = A[t, Nt^{-1}] \subset A[t, t^{-1}].$$

These algebras glue and define a flat family

$$\mathfrak{D}_R \longrightarrow \mathbb{A}^1.$$

Over $\mathbb{G}_m$ it is $\widehat{D}_R \times \mathbb{G}_m$, while its special fibre is

$$\mathfrak{D}_{R,0} \simeq \text{Spec} \widehat{\Delta}_R \text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = C_{\widehat{\Delta}_R/\widehat{D}_R}.$$

The construction is functorial for isomorphisms of the abstract failure scheme.

Proof. The extended Rees algebra is the standard construction. Multiplication by t is injective because $\mathcal{R}_N$ is a subring of $A[t, t^{-1}]$. Hence it is torsion-free over the PID $\mathbb{C}[t]$, and therefore flat. Inverting t gives $A[t, t^{-1}]$. Modulo t , the class of nt^{-j} , $n \in N^j$, depends only on $n \bmod N^{j+1}$, giving

$$\mathcal{R}_N/(t) \simeq \bigoplus_{j \geq 0} N^j/N^{j+1}.$$

Localization compatibility glues the construction. Since N is the ideal of the reduction $\widehat{\Delta}_R$ in $\widehat{D}_R$, the special fibre is the normal cone. The nilradical is intrinsic, so the construction is functorial. $\square$

The point of Revision 128 is not Proposition 12.1 itself. The new content is that its special fibre is now bounded and stratified:

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j=1}^5 \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

with the quotient multiplication (6.4). Projection corank two has the mixed-kernel atlas of Theorem 6.4; all A_H -rank drops have the finite valuation bounds of Proposition 6.8; projection corank three has the exact index-five/index-six dichotomy of Theorem 6.9; and projection corank four has exact index five by Theorem 6.10.

Flattening is used only to organize coefficient specializations of these explicitly presented finite modules. We use the standard flattening-stratification formalism in [20, Tag 052F]; no novelty is claimed for that formalism either.

Remark 12.2 (Novelty boundary). The standard ingredients in this section are the extended Rees algebra, deformation to the normal cone, and flattening stratification. The manuscript-specific statements are the Fitting factorization, the global determinant-power bound $d^5 \in J$, the nine mixed-kernel normal forms and their graded supports, the corank-three Schur-type exclusion, the corank-four Jacobian-covariant calculation, the collision primary laws, and the polarized reconstruction theorem. These roles are kept separate in the statements and in the referee response.

13. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem 1.4. Work on the standard chart of $\mathrm{Gr}(4, \mathrm{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0x_1, x_0x_2, x_0x_3, x_1x_2, x_1x_3, x_2x_3).$$

For $C \in \mathrm{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (13.1)$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma 13.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (13.2)$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\mathrm{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\mathrm{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \mathrm{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (13.3)$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (13.4)$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix B specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C^*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$. Every monomial of those degrees belongs to the respective homogeneous ideal over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition 13.2. *The Jacobian map on G_4° has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma 13.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition 2.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem 1.4. Combine Proposition 13.2 with the recovery of the polarized quartic in Theorem 1.1 and the algebra-orbit equivalence of Lemma 2.4. $\square$

Corollary 13.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.*

Proof. Flatness is Corollary 3.6. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem 1.1. The graph description of Δ in (3.2) is independent of R . $\square$

14. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by

$q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (14.1)$$

Proposition 14.1. *For $R \in G_4^\circ$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem 1.1 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth. Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (14.2)$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition 14.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

15. REYE GEOMETRY, INFINITESIMAL MODULI, AND THE NOVELTY BOUNDARY

We now separate three statements that were too close together in the preceding revision: the classical moduli of webs and Reye congruences, the exact differential calculation for the polarized K3 map, and the nonreduced reconstruction theorem.

15.1. The classical nine-dimensional quotient. For $e = 4$, a relation space $R \subset \text{Sym}^2 V$ is a web of quadrics. The bilinear incidence surface Z_R of §14 is a K3 surface with its fixed-point-free factor-exchange involution; its quotient is the classical Reye Enriques surface associated with the web. This construction and the resulting congruence of lines are classical; see [7, 1, 10, 13, 9, 16].

Arrondo–Sols compute that the Hilbert scheme of Reye congruences is smooth of dimension 24 and rational, and describe it by an open Grassmannian parameter of the corresponding data. After quotienting by the 15-dimensional projective coordinate group, the expected moduli dimension is therefore nine. In characteristic zero the nodal Enriques locus is the classical nine-dimensional divisor in the ten-dimensional Enriques moduli space; the modern theorem [16] identifies every classical nodal Enriques surface as a Reye congruence.

Thus the equality

$$24 - 15 = 9$$

is not a novelty claim of this paper.

Proposition 15.1 (The classical tangent quotient). *At a general regular web with finite projective stabilizer, the tangent space to the Reye moduli locus is the quotient of the 24-dimensional web tangent space by the 15-dimensional infinitesimal $\mathrm{PGL}(V)$ -orbit:*

$$T_{[R]}\mathcal{M}_{\mathrm{Reye}} \simeq T_R \mathrm{Gr}(4, \mathrm{Sym}^2 V) / \mathfrak{pgl}(V), \quad \dim T_{[R]}\mathcal{M}_{\mathrm{Reye}} = 9.$$

The differential of the polarized-K3 map of Theorem 1.4 factors through this quotient.

Proof. The first statement is the tangent-space form of the smooth 24-dimensional Hilbert calculation of Arrondo–Sols on the regular web locus, followed by the infinitesimal quotient by projective coordinates. The stabilizer is finite at a general point, so the orbit tangent has dimension 15.

The construction $R \mapsto Z_R \simeq Y_R$ is equivariant for $\mathrm{PGL}(V)$. Passing from an embedded quartic to its polarized isomorphism class kills the infinitesimal projective orbit; hence the differential factors through the displayed quotient. $\square$

Proposition 15.2 (Web parameter and Reye Hilbert point). *Let*

$$\mathcal{G} = \mathrm{Gr}(4, \mathrm{Sym}^2 V)$$

and let $\mathcal{G}^{\mathrm{reg}}$ be the open on which the classical Reye construction has the expected Hilbert polynomial. The assignment of a web to its Reye congruence is a $\mathrm{PGL}(V)$ -equivariant morphism

$$\Phi : \mathcal{G}^{\mathrm{reg}} \longrightarrow \mathrm{Hilb}_{\mathrm{Reye}}.$$

Under the Grassmannian parameterization of the regular Reye locus in Arrondo–Sols, Φ identifies a dense open of $\mathcal{G}^{\mathrm{reg}}$ with the corresponding 24-dimensional Hilbert component. In particular, at a general regular web,

$$d\Phi_R : \mathrm{Hom}(R, \mathrm{Sym}^2 V/R) \xrightarrow{\sim} T_{[\mathrm{Reye}(R)]} \mathrm{Hilb}_{\mathrm{Reye}}.$$

After quotienting by the infinitesimal projective orbit this gives the tangent quotient in Proposition 15.1.

Proof. Over $\mathcal{G}$ the universal rank-four subbundle of $\mathrm{Sym}^2 V$ gives the universal web. The incidence equations defining the Reye congruence are polynomial in the universal Pluecker coordinates. Restrict to the open on which their Hilbert polynomial is the regular Reye polynomial. Flatness there and the Hilbert functor give the morphism Φ ; equivariance is immediate from the construction.

Arrondo–Sols describe the regular Reye Hilbert component by the open Grassmannian parameter of the defining four-dimensional system of quadrics. With the present convention this parameter is precisely a dense open of $\mathrm{Gr}(4, \mathrm{Sym}^2 V)$ (duality only changes whether one writes the system or its annihilator). Their smoothness and dimension-24 statement therefore identifies the tangent space of the Hilbert point with the tangent space of this Grassmannian. The latter is $\mathrm{Hom}(R, \mathrm{Sym}^2 V/R)$, proving the displayed isomorphism. Since the construction is $\mathrm{PGL}(V)$ -equivariant, the orbit tangent maps to the orbit tangent and the quotient gives Proposition 15.1. $\square$

The role of Lemma 13.1 is now precise. It is not used to discover the number nine. Its exact 25×25 integral minor proves that the particular polarized K3 attached to our multiplication problem does not collapse any of the 24 embedded web directions at the explicit rational point. After projective quotient, it proves rank nine for the map to polarized K3 moduli. Since maximal rank is open, the induced map from the classical nine-dimensional Reye deformation space to our polarized K3 image is generically unramified. This is the conceptual tangent statement behind Theorem 1.4; the determinant is an exact arithmetic witness for it.

15.2. What the finite ambiguity means. The preceding paragraph also locates the remaining Torelli ambiguity. The failure scheme first reconstructs the polarized K3 $(Y_R, \mathcal{O}_{Y_R}(1))$. A relation system additionally supplies the free involution coming from factor exchange on the bilinear incidence model and hence an Enriques quotient with its Reye data. Classical Reye geometry then has substantial rigidity. In particular, Dolgachev–Reider prove that for a Reye polarization on an Enriques surface the stable rank-two bundle with c_1 equal to that polarization and $c_2 = 3$ is the Reye bundle and is unique up to isomorphism; it is the restriction of the universal quotient bundle from $G(2, 4)$.

Consequently the generic finite fibre in Theorem 1.4 should not be read as an uncontrolled family of quadratic tensors. Once a compatible Enriques quotient and Reye polarization are fixed, the Grassmannian Reye bundle is rigid in the preceding sense. The remaining finite packet is concentrated in the finite choices of compatible Reye/Enriques data over the recovered polarized K3, together with the passage back to the web. The present paper proves finiteness but does not claim a numerical degree or generic degree one; such a claim would require a separate enumeration of those compatible involutions and web structures.

There is a second useful localization. On the mixed-regular projection-corank-two locus, the unsplit contact quartic of §8 is the restriction of the already recovered quartic Y_R to the pencil of hyperplanes containing H . Thus the generic corank-two contact cover itself is determined by the polarized K3. The following theorem turns the finite Torelli packet into a precise finite-cover statement and identifies its monodromy.

Theorem 15.3 (Generic Torelli packet as a finite étale monodromy cover). *Let $\mathcal{W}$ be the irreducible nine-dimensional coarse quotient of the regular web locus by projective equivalence on the finite-stabilizer open, and let $\mathcal{K}$ be the nine-dimensional polarized-K3 image of Theorem 1.4. There are nonempty opens*

$$\mathcal{W}^\circ \subset \mathcal{W}, \quad \mathcal{K}^\circ \subset \mathcal{K}$$

such that

$$\tau : \mathcal{W}^\circ \longrightarrow \mathcal{K}^\circ$$

is finite étale of a constant degree

$$d_{\text{Tor}} = [\mathbb{C}(\mathcal{W}) : \mathbb{C}(\mathcal{K})].$$

Its geometric monodromy is transitive on the d_{Tor} points of a fibre. Equivalently, if L is the normal closure of $\mathbb{C}(\mathcal{W})/\mathbb{C}(\mathcal{K})$, the monodromy group is the permutation image of

$$\text{Gal}(L/\mathbb{C}(\mathcal{K}))$$

on the embeddings of $\mathbb{C}(\mathcal{W})$ in L .

The degree-four contact incidence and its discriminant are pulled back from the polarized quartic on $\mathcal{K}^\circ$. Hence they are constant on a Torelli fibre and do not account for d_{Tor} ; any further separation of the finite packet must use web-dependent structure beyond the polarized quartic, such as the exceptional mixed-kernel or deeper projection-corank strata.

Proof. The source $\mathcal{W}$ is irreducible because it is obtained from a nonempty open of the irreducible Grassmannian $\text{Gr}(4, \text{Sym}^2 V)$ by the projective-coordinate quotient on the finite-stabilizer locus. Theorem 1.4 says that the dominant map $\tau : \mathcal{W} \dashrightarrow \mathcal{K}$ is generically finite. Thus the induced extension of function fields is finite. In characteristic zero it is separable.

Normalize a nonempty affine open of $\mathcal{K}$ in $\mathbb{C}(\mathcal{W})$. After deleting the complement of the locus on which the original rational map is defined, and then deleting the branch and nonflat loci, one obtains a finite étale morphism from a nonempty open of $\mathcal{W}$. Its degree is the function-field degree displayed above. Connectedness of the generic source, or equivalently irreducibility of $\mathcal{W}$,

gives transitivity of geometric monodromy. The description by the normal closure is the standard field-theoretic realization of that action.

The contact incidence is defined solely from the recovered embedded quartic $Y \subset \mathbb{P}^3$ by

$$(H, [\alpha]) \quad \text{with} \quad H \subset \ker \alpha,$$

and the discriminant is the discriminant of its restricted binary quartic. Both constructions commute with isomorphism of the polarized quartic. They therefore descend to $\mathcal{K}^\circ$ and pull back along τ , proving the last assertion. $\square$

15.3. Comparison with nonreduced reconstruction theory. The use of nilpotent layers has classical antecedents. A ribbon is a square-zero double structure with an invertible conormal module, and Bayer–Eisenbud developed its intrinsic and canonical geometry [4]. Manolache surveys Cohen–Macaulay nilpotent structures and their filtrations [15]. Primitive multiple schemes carry a canonical filtration by lower multiplicities in the sense of Drézet [11].

Our local corank-one model

$$t(t, f) = (t) \cap (t, f)^2$$

indeed has a square-zero nilradical line on the embedded support, and the corank-two model has one further nonzero nilpotent power. We do not claim that the existence of a nilpotent filtration is new. The specific contribution is the following chain of implications, which does not occur in the cited multiple-structure theory:

$$\text{multiplication Fitting scheme} \implies (\mathcal{Z}_1, \widehat{\mathcal{N}}|_{\mathcal{Z}_1}) \implies \omega_{\mathcal{Z}_1} \otimes \widehat{\mathcal{N}}^{-9} \implies (Y_R, \mathcal{O}_{Y_R}(1)),$$

together with

$$\mathcal{Z}_2 = \text{rank-two Schubert support}$$

on the mixed-regular higher-corank locus. The ambient failure scheme is not assumed to be a primitive multiple scheme, and the filtration is defined by annihilators of powers of its intrinsic nilradical rather than by choosing an ambient thickening.

15.4. Ballico’s failure-locus framework. The complete theorem text of Ballico’s 1993 article [2] was not available to this revision. We therefore do not infer either anticipation or non-anticipation from its title or metadata. A theorem-level comparison is possible with Ballico’s accessible 1996 continuation [3]. There the basic data are an integral projective variety X , a very ample line bundle L , the complete embedding by $H^0(X, L)$, and multiplication maps such as

$$H^0(L) \otimes H^0(L^k) \longrightarrow H^0(L^{k+1}).$$

The principal results propagate failure of quadratic or higher normality to suitable linear sections and finite subschemes; the parameter Grassmannian is a Grassmannian of projective linear sections.

The present construction has a different input and a different scheme-theoretic output. The fixed object is the finite local cube-zero algebra B_R ; a point of X_R is a unital subspace $\mathbb{C}1 \oplus K \subset B_R$; and the failure scheme is the zeroth Fitting scheme of

$$\text{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}.$$

Its embedded associated supports, nilradical powers, and the polarized K3 reconstructed from their line bundles are part of the statement. Ballico’s 1996 theorems are therefore an important precedent for organizing multiplication failure over a Grassmannian, but they do not by themselves supply the Fitting normal forms or the nilpotent-depth reconstruction proved here. This comparison is deliberately limited to the theorem text actually inspected; the separate 1993 comparison remains documentary work rather than a premise in a novelty claim.

15.5. Residue of the main theorem. After these classical comparisons, the claims that remain specific to this paper are concise.

First, an abstract multiplication-failure scheme reconstructs the quartic polarization by the intrinsic line $\omega_{E_R} \otimes \mathcal{N}_R^{-9}$ and its section ring. Second, the same nonreduced scheme has a functorial annihilator filtration whose next nonzero stage recovers the first higher-corank Schubert support. Third, on the mixed-regular rank-two locus the unsplit contact ideal has a complete collision specialization law, the fifth-power vertex component remains genuinely primary, and the associated graded nilpotent algebra identifies both the contact cone and the rank-two Schubert support. The first decomposable mixed-kernel wall is also computed scheme-theoretically. Finally, Theorem 15.3 identifies the generic inverse problem with a finite étale monodromy cover. These statements are independent of whether the nine-dimensional Reye family itself is classical.

APPENDIX A. WEIGHTED DETERMINANTAL FLAGS AND THE CLASSICAL ORDER PROFILE

We record explicitly how the retained weighted-matrix calculation fits the classical determinantal theory. This comparison concerns the generic matrix only; the fixed-tensor theorem of §7 was proved independently.

Let $A = (a_{ij})$ be a generic $q \times p$ matrix over a characteristic-zero field k , with $p \geq q$, and put $R_0 = k[a_{ij}]$, $I_0(A) = R_0$. For a nonincreasing sequence $\rho = (r_1, \dots, r_s)$ write

$$I^\rho = \prod_{\nu=1}^s I_{r_\nu}(A), \quad \gamma_j(\rho) = \sum_{\nu=1}^s \max(r_\nu - j + 1, 0).$$

The classical formula is

$$I^\rho = \bigcap_{j=1}^{r_1} I_j(A)^{(\gamma_j(\rho))}. \quad (\text{A.1})$$

Here the superscript in parentheses denotes symbolic power. See De Concini–Eisenbud–Procesi [8], and its statement as Theorem 2.4 in Bruns–Conca [5], together with Proposition 2.2 there for the symbolic-order basis. The characteristic hypotheses and attribution are stated on printed pages 13–14 of that exposition.

Proposition A.1 (Coefficientwise suspension). *In $R_0[t]$, put*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (\text{A.2})$$

This is an irredundant primary decomposition. At a point with $t = 0$ where A has rank a , the nilradical of the corresponding local quotient has index $q - a + 1$.

Proof. In the regular local ring $(R_0)_{I_j}$, the symbolic order of an i -minor is $\max(i - j + 1, 0)$. Adjoining t gives the coefficient rule

$$\sum_{u \geq 0} t^u f_u \in P_j^{(b)} \iff f_u \in I_j^{(\max(b-u, 0))} \text{ for every } u. \quad (\text{A.3})$$

For precision, the associated graded ring of the maximal ideal $(t, I_j(R_0)_{I_j})$ is a polynomial ring over its residue field; its homogeneous t -degrees are independent. The order of the displayed sum is therefore the minimum of $u + \text{ord}_{I_j}(f_u)$. Contracting the maximal-ideal power proves (A.3), including the assertion that each contracted symbolic power is primary.

Intersecting with (t) deletes coefficient degree zero. For $1 \leq u \leq q$, put $k = q + 1 - u$. The remaining coefficient conditions on the right of (A.2) are

$$f_u \in \bigcap_{j=1}^k I_j^{(k-j+1)} = I_k,$$

the one-factor case $\rho = (k)$ of (A.1). For $u \geq q + 1$ there is no condition. These are exactly the coefficients of F , proving the equality.

For irredundancy, let d_{j-1} be a fixed $(j-1)$ -minor, with $d_0 = 1$. The element $t^{q-j+1}d_{j-1}^2$ belongs to every component other than $P_j^{(q-j+2)}$. Indeed, its order along P_j is $q-j+1$, while at P_k for $k < j$ its order is $q-j+1+2(j-k) \geq q-k+2$, and for $k > j$ its t -order suffices. The element d_q^2 witnesses the necessity of (t) . The associated primes are therefore exactly the displayed radicals.

After inverting an a -minor and Schur reducing, the localized coefficient ideal contains t^{q-a+1} , whereas t^{q-a} has coefficient one in degree $q-a$, where membership would require a one-minor of the residual matrix. That coefficient ideal is proper at the point. Thus the radical (t) has the asserted exact index. $\square$

The exponent $q-j+2$ is therefore the scalar-variable suspension of the classical minor order $k-j+1$, taken coefficientwise. This is a direct consequence of the classical profile plus (A.3); it is not a new straightening theorem. The presentation $I_q[A, tI_q] = \sum_i t^{q-i}I_i(A)$ can also be viewed as substituting a scalar identity block in a larger generic matrix. Such substitution alone does not justify preservation of primary components or flatness. The coefficient proof is what supplies the exact intersection here, while the ideal $\delta\mathfrak{m} + (g_0, \dots, g_4)$ in §7 requires its separate fixed-tensor analysis.

APPENDIX B. AN EXPLICIT SYSTEM AND EXACT CERTIFICATE SPECIFICATIONS

The rational system used throughout is the following. It is the same relation system as in the preceding manuscript's exact smoothness example, now used for a differential and a higher-corank calculation.

$$\begin{aligned} q_0 &= x_0^2 + 2x_0x_1 - 2x_0x_2 + 2x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3 \\ q_1 &= -x_0x_1 - x_0x_3 + x_1^2 - x_1x_2 - x_1x_3 - 2x_2x_3 \\ q_2 &= 2x_0x_1 - x_0x_2 - 2x_0x_3 + 2x_1x_2 + x_2^2 + x_2x_3 \\ q_3 &= x_0x_2 + x_1x_2 - 2x_2x_3 + x_3^2 \end{aligned}$$

For a compact complete specification of its quartic, list the 35 degree-four exponent tuples in descending lexicographic order, starting with $(4, 0, 0, 0)$ and ending with $(0, 0, 0, 4)$. The coefficients of $F = \det(\partial q_i / \partial x_j)$ in that order are the rows of

$$\begin{pmatrix} -8 & -4 & -20 & 16 & 20 & 0 & -28 \\ -2 & -36 & 40 & 20 & -50 & -20 & -26 \\ 72 & -6 & 0 & -66 & -6 & 30 & 4 \\ -14 & -4 & -6 & -12 & 28 & 18 & -56 \\ 28 & 6 & -18 & 8 & -2 & 16 & 2 \end{pmatrix}.$$

Thus this display, together with the fixed monomial order, specifies every coefficient, including zeros. Alternatively, the determinant of the displayed four quadrics is its coordinate-free recipe.

For any list of homogeneous forms $f_1, \dots, f_s$ define the degree- d Macaulay coefficient matrix $\mathcal{M}_d(f)$ as follows. Rows are all degree- d monomials in the same descending lexicographic order. Columns are the forms $x^\beta f_i$, with i outermost and all β of degree $d - \deg f_i$ in descending

lexicographic order innermost. The entry in row α is the coefficient of x^α . All entries for this example are integers.

Matrix	Size	Certified rank	Minor modulo 101
$[F, G_{01}, \dots, G_{36}]$	35×25	25	41
$\mathcal{M}_5(q_0, q_1, q_2, q_3)$	56×80	56	44
$\mathcal{M}_9(F_{x_0}, F_{x_1}, F_{x_2}, F_{x_3})$	220×336	220	79

The differential minor consists of the first 25 rows and all columns; its full integer value is (13.4). The following deterministic rule specifies the other two minors without printing large matrices. Reduce the displayed coefficient matrix modulo 101. Scan columns from left to right. At the next pivot step choose the first remaining row with nonzero entry, exchange it with the next pivot row, normalize that row, and eliminate below. Skip a column if there is no nonzero remaining entry. Record the original row indices in the order selected and the pivot columns. The square submatrix of the *original*, unreduced matrix with exactly these row and column orders has the residue in the final column of the table. Full recorded index lists, along with the coefficient-generating program, are included in `evidence/K3_CERTIFICATES.json` and `checks/exact_k3.py`.

Every step is finite-field arithmetic on integer data. A nonzero residue of a specified square minor proves that integer minor is nonzero, so it proves a rank statement over $\mathbb{Q}$, not merely a rank in one numerical specialization. The Macaulay rank statements imply the homogeneous ideal containments used in Lemma 13.1. The exact integer determinant is also recomputed independently with rational-domain elimination. No floating-point rank, tolerance, or unverified smoothness inference is used.

For the corank-two quotient in (7.1), one may reproduce its quotient map by multiplying the six-dimensional quadratic quotient by

$$\begin{pmatrix} 0 & -1/2 & -1 & 1 & 0 & 0 \\ 0 & -1/2 & -1 & 0 & 1 & 0 \\ 0 & -3/2 & -2 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix annihilates the three columns of $\text{Sym}^2 H$ and has rank three. It yields the displayed B_H, C_H and the quartic (7.2) by ordinary multiplication. The factor 2 in the coefficient of uv is essential: it records the square of $ux_2 + vx_3$, not a divided-power convention.

An additional exact regression test uses the normalized matrix (7.3) with

$$C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & -3 \end{pmatrix}.$$

Then $h(u, v) = uv(u-v)(u-2v)$, so the four ideals are (a, b) , (c, d) , $(a-c, b-d)$, and $(a-2c, b-2d)$. Elimination computes both sides of (1.4) over $\mathbb{Q}$, and mutual Gröbner reduction verifies their equality. It also checks that δ^2 is nonzero, that every element of $\mathfrak{m}\delta^2$ vanishes, and that $\delta^3 = 0$. This test checks a concrete instance; the general coefficient-ring proof is Lemma 7.4 and Theorem 7.5, not an extrapolation from that instance.

REFERENCES

- [1] E. Arrondo and I. Sols, *On congruences of lines in the projective space* (Chapter 6 written in collaboration with M. Pedreira), *Mém. Soc. Math. France (N.S.)* **50** (1992), 1–96. doi:10.24033/msmf.363.
- [2] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [3] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.

- [4] D. Bayer and D. Eisenbud, Ribbons and their canonical embeddings, *Trans. Amer. Math. Soc.* **347** (1995), 719–756. doi:10.1090/S0002-9947-1995-1273472-3.
- [5] W. Bruns and A. Conca, Gröbner bases and determinantal ideals, in *Commutative Algebra, Singularities and Computer Algebra*, NATO Sci. Ser. II Math. Phys. Chem. 115, Kluwer, 2003, 9–66; author preprint arXiv:math/0302058.
- [6] W. Bruns and U. Vetter, *Determinantal Rings*, Lecture Notes in Mathematics 1327, Springer, 1988.
- [7] F. Cossec, Reye congruences, *Trans. Amer. Math. Soc.* **280** (1983), 737–751. doi:10.1090/S0002-9947-1983-0716848-4.
- [8] C. De Concini, D. Eisenbud, and C. Procesi, Young diagrams and determinantal varieties, *Invent. Math.* **56** (1980), 129–165. doi:10.1007/BF01392548.
- [9] I. Dolgachev and S. Kondō, *Enriques Surfaces II*, Springer, 2025, chapter *Reye Congruences*, 67–141. doi:10.1007/978-981-96-1513-1_2.
- [10] I. Dolgachev and I. Reider, On rank 2 vector bundles with $c_1^2 = 10$ and $c_2 = 3$ on Enriques surfaces, in *Algebraic Geometry*, Lecture Notes in Mathematics 1479, Springer, 1991, 39–49. doi:10.1007/BFb0086262.
- [11] J.-M. Drézet, Primitive multiple schemes, *Eur. J. Math.* **7** (2021), 985–1045. doi:10.1007/s40879-020-00447-4.
- [12] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics 158, Cambridge University Press, 2016. doi:10.1017/CBO9781316594193.
- [13] C. Ingalls and A. Kuznetsov, On nodal Enriques surfaces and quartic double solids, *Math. Ann.* **361** (2015), 107–133. doi:10.1007/s00208-014-1066-y.
- [14] J. Kollár, *Rational Curves on Algebraic Varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 32, Springer, 1996, Chapter IV, §5.
- [15] N. Manolache, Cohen–Macaulay nilpotent schemes, arXiv:math/0312514 (2003).
- [16] G. Martin, G. Mezzedimi, and D. C. Veniani, Nodal Enriques surfaces are Reye congruences, *J. Reine Angew. Math.* **808** (2024), 49–65. doi:10.1515/crelle-2023-0092.
- [17] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 2, Springer, 1998, §5.1.
- [18] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [19] The Stacks Project Authors, *The Stacks Project*, Section 31.20, “The normal cone of an immersion,” Tag 062Z, <https://stacks.math.columbia.edu/tag/062Z>.
- [20] The Stacks Project Authors, *The Stacks Project*, Section 38.21, “Flattening stratifications,” Tag 052F, <https://stacks.math.columbia.edu/tag/052F>.
- [21] The Stacks Project Authors, *The Stacks Project*, Section 31.7, “Relative assassin,” Tag 05AS, <https://stacks.math.columbia.edu/tag/05AS>.
- [22] The Stacks Project Authors, *The Stacks Project*, Section 37.18, “Flat modules and relative assassins,” Tag 0GSF; in particular Lemma 37.18.5, Tag 0GSJ.
- [23] The Stacks Project Authors, *The Stacks Project*, Section 38.18, “A criterion for purity,” Tag 05L2.
- [24] The Stacks Project Authors, *The Stacks Project*, Section 37.25, “Relative assassins,” in particular Lemma 37.25.5, Tag 05KR, <https://stacks.math.columbia.edu/tag/05KR>.